

PCORnet Empirical Data Curation Report Table of Contents

This report is derived from the results of the data curation query. This query package fully characterizes all CDM v7.0 tables. Please refer to the PCORnet Empirical Data Curation Report v7.01 Specifications for additional details about this report.

Report Run Date: 2026-04-09

Query Run Date: 2026-04-08

Maximum Table Refresh Date: 2026-04-06

Query Package: DC V7.01

Lookback Date: 2016-04-08

CDM Version: 070

SAS_ETS licensed: Yes / SAS_ETS installed: Yes

Section	Table	Table Description	Data Check(s)
Data Check Summary	DC Summary	Data Check Exception Summary	n/a
Section I: Descriptive Information	Table IA	Demographic Summary	n/a
	Table IB	Potential Pools of Patients	2.09, 3.04, 3.05
	Table IC	Height, Weight, and Body Mass Index (BMI)	n/a
	Table ID	Records, Refresh Dates, Patients, Encounters, and Date Ranges by Table	1.18
	Table IE	Records Per Table by Encounter Type	n/a
	Table IF	Date Obfuscation or Imputation	n/a
	Chart IA	Trend in Vital Measures by Measurement Date, Past 5 Years	n/a
	Chart IB	Trend in Encounters by Admit Date and Encounter Type, Past 5 Years	n/a
	Chart IC	Trend in Institutional Encounters by Discharge Date and Encounter Type, Past 5 Years	n/a
	Chart ID	Trend in Laboratory Results by Result Date, Past 5 Years	n/a
	Chart IE	Trend in Prescribed Medications by Rx Order Date, Past 5 Years	n/a
	Chart IF	Trend in Dispensed Medications by Dispense Date, Past 5 Years	n/a
	Chart IG	Trend in Administered Medications by Start Date, Past 5 Years	n/a
	Chart IH	Trend in Condition Records by Report Date, Past 5 Years	n/a
	Chart II	Trend in Death Records by Death Date and Source, Past 5 Years	n/a
	Chart IJ	Trend in Immunization Records by Vx Record Date, Past 5 Years	n/a
	Chart IK	Trend in Clinical Observation Records by Start Date, Past 5 Years	n/a
	Chart IL	Trend in General Observation Records by Start Date, Past 5 Years	n/a

REPORT_RUN_DATE is the date the EDC report was run. QUERY_RUN_DATE is the date the data curation query (Part 1 portion if running the program in parts) was run. MAXIMUM_TABLE_REFRESH_DATE is derived from the HARVEST table. The LOOKBACK_DATE is the earliest date eligible for inclusion in the query results (see the Work Plan for details). CDM_VERSION is from the HARVEST table. The SAS_ETS module is used for Data Check 2.08.

PCORnet Empirical Data Curation Report Table of Contents (continued - page 2 of 3)

This report is derived from the results of the data curation query. This query package fully characterizes all CDM v7.0 tables. Please refer to the PCORnet Empirical Data Curation Report v7.01 Specifications for additional details about this report.

Section	Table	Table Description	Data Check(s)
Section II: Data Model Conformance	Table IIA	Primary Key Errors	1.05
	Table IIB	Values Outside of Common Data Model (CDM) Specifications	1.06
	Table IIC	Non-Permissible Missing Values	1.07
	Table IID	Diagnostic Errors	1.01, 1.02, 1.03, 1.04, 1.15, 1.17, 1.21
	Table IIE	Orphan Records, Replication Errors, Encounter Duplication and Hash Token Duplication	1.08, 1.09, 1.10, 1.11, 1.12, 1.14, 1.19
	Table IIF	Potential Code Errors and Misplaced Codes	1.13, 1.16
	Table IIG	LOINC Panel Codes	1.20
	Table IIIA	Future Dates	2.01
	Table IIIB	Records with Extreme Values	2.02
	Table IIIC	Illogical Dates	2.03
Section III: Data Plausibility	Table IIID	Encounters Per Visit and Per Patient	2.04
	Table IIIG	Monthly Record Volume Outliers, Selected Domains	2.08
	Chart IIIA	Monthly Record Volume Outliers, Encounters	2.08
	Chart IIIB	Monthly Record Volume Outliers, Diagnoses	2.08
	Chart IIIC	Monthly Record Volume Outliers, Procedures	2.08
	Chart IIID	Monthly Record Volume Outliers, Vitals	2.08
	Chart IIIE	Monthly Record Volume Outliers, Prescribing	2.08
	Chart IIIF	Monthly Record Volume Outliers, Labs	2.08
	Chart IIIG	Monthly Record Volume Outliers, Med Admin	2.08

REPORT_RUN_DATE is the date the EDC report was run. QUERY_RUN_DATE is the date the data curation query (Part 1 portion if running the program in parts) was run. MAXIMUM_TABLE_REFRESH_DATE is derived from the HARVEST table. The LOOKBACK_DATE is the earliest date eligible for inclusion in the query results (see the Work Plan for details). CDM_VERSION is from the HARVEST table. The SAS_ETS module is used for Data Check 2.08.

PCORnet Empirical Data Curation Report Table of Contents (continued - page 3 of 3)

This report is derived from the results of the data curation query. This query package fully characterizes all CDM v7.0 tables. Please refer to the PCORnet Empirical Data Curation Report v7.01 Specifications for additional details about this report.

Section	Table	Table Description	Data Check(s)
Section IV: Data Completeness and Plausibility	Table IVA	Diagnosis Records Per Encounter, Overall and by Encounter Type	3.01
	Chart IVA	Diagnosis Records Per Encounter by Admit Date and Encounter Type, Past 5 Years	n/a
	Table IVB	Procedure Records Per Encounter, Overall and by Encounter Type	3.02
	Chart IVB	Procedure Records Per Encounter by Admit Date and Encounter Type, Past 5 Years	n/a
	Table IVC	Missing or Unknown Values, Required Tables	3.03
	Table IVD	Missing or Unknown Values, Optional Tables	3.03
	Table IVE	Principal Diagnoses for Institutional Encounters	2.07, 3.06
	Table IVF	Data Latency and Completeness of Encounter, Diagnosis and Procedure Data, Past 2 Years	3.07
	Table IVG	Data Latency and Completeness of Vital, Prescription, and Lab Data, Past 2 Years	3.11
	Table IVH	RXNORM Term Type Mapping	3.08, 3.15
	Table IVI	Laboratory and Clinical Observation Result Data Completeness	3.09, 3.10, 3.12, 3.16, 3.17
	Table IVI_Ref	Laboratory and Clinical Observation Result Data Completeness Definitions	n/a
	Table IVJ	Data Latency and Completeness of Medication Administration, Dispensing and Clinical Observation Data, Past 2 Years	3.14
Section V: Data Persistence	Table VA	Changes in Tables	4.01
	Table VB	Changes in Selected Encounter Types and Domains	4.02
	Table VC	Changes in Selected Code Types	4.03

REPORT_RUN_DATE is the date the EDC report was run. QUERY_RUN_DATE is the date the data curation query (Part 1 portion if running the program in parts) was run. MAXIMUM_TABLE_REFRESH_DATE is derived from the HARVEST table. The LOOKBACK_DATE is the earliest date eligible for inclusion in the query results (see the Work Plan for details). CDM_VERSION is from the HARVEST table. The SAS_ETC module is used for Data Check 2.08.

Data Check Exception Summary

This table summarizes conformance with PCORnet Data Checks v19. Changes to the previous cycle's data checks are explained in the footnotes. Exceptions to required data checks are highlighted in red and must be corrected before returning results. Exceptions to investigative data checks are highlighted in blue and must be investigated and explained in the ETL ADD. Network-wide results for PCORnet DataMarts approved as of November 3, 2025 are presented for comparison; a result of 10.3 means that 10.3% of DataMarts had exceptions to this data check.

Data Check	EDC Table	Data Check Description	Category	Type	Exception?	% DataMarts w/ exceptions
DC 1.01 ^a	Table IID	Required tables are not present	Data Model Conformance	Required	No	0
DC 1.02 ^a	Table IID	Required tables are not populated. DEMOGRAPHIC, ENROLLMENT, ENCOUNTER, DIAGNOSIS, PROCEDURES, and HARVEST are required for all network partners; LAB_RESULT_CM, PRESCRIBING, and VITAL are required for network partners with electronic health record data	Data Model Conformance	Required	No	0
DC 1.03 ^a	Table IID	Required fields are not present	Data Model Conformance	Required	No	0
DC 1.04 ^a	Table IID	Required fields do not conform to data model specifications for data type, length, or name	Data Model Conformance	Required	No	0
DC 1.05	Table IIA	Tables have primary key definition errors	Data Model Conformance	Required	No	0
DC 1.06 ^b	Table IIB	Required fields contain values outside of data model specifications	Data Model Conformance	Required	No	0
DC 1.07	Table IIC	Required fields have non-permissible missing values	Data Model Conformance	Required	No	0
DC 1.08	Table IIE	Tables contain orphan PATIDS (PATIDS that are not present in the DEMOGRAPHIC table)	Data Model Conformance	Required	No	0
DC 1.09	Table IIE	Tables contain orphan ENCOUNTERIDs (ENCOUNTERIDs that are not present in the ENCOUNTER table) for more than 5% of records	Data Model Conformance	Required	No	0
DC 1.10	Table IIE	Replication errors between the ENCOUNTER, PROCEDURES and DIAGNOSIS tables. Replication errors are ENCOUNTERIDs in the DIAGNOSIS or PROCEDURES table where the encounter type or admit date does not match the corresponding value in the ENCOUNTER table	Data Model Conformance	Required	No	0
DC 1.11	Table IIE	More than 5% of encounters are assigned to more than one patient	Data Model Conformance	Required	No	0
DC 1.12	Table IIE	Tables contain orphan PROVIDERIDs (PROVIDERIDs that are not present in the PROVIDER table)	Data Model Conformance	Required	No	0
DC 1.13 ^c	Table IIF	More than 5% of CPT®/HCPCS, CVX, ICD, NDC, LOINC or RXNORM codes do not conform to the expected length or content based on terminology-specific heuristics	Data Model Conformance	Required	No	0
DC 1.14	Table IIE	Patients in the DEMOGRAPHIC table are not in the HASH_TOKEN table	Data Model Conformance	Investigative	No	13

(a) Can be detected by CDM Diagnostic Query.

(b) Non-conforming values in most fields can be detected by the Value Set Conformance Query

(c) Can be detected by the Potential Code Errors Query

Data Check Exception Summary (continued - page 2 of 6)

This table summarizes conformance with PCORnet Data Checks v19. Changes to the previous cycle's data checks are explained in the footnotes. Exceptions to required data checks are highlighted in red and must be corrected before returning results. Exceptions to investigative data checks are highlighted in blue and must be investigated and explained in the ETL ADD. Network-wide results for PCORnet DataMarts approved as of November 3, 2025 are presented for comparison; a result of 10.3 means that 10.3% of DataMarts had exceptions to this data check.

Data Check	EDC Table	Data Check Description	Category	Type	Exception?	% DataMarts w/ exceptions
DC 1.15 ^a	Table IID	Fields with undefined lengths that are present in more than one table (PATID, ENCOUNTERID, PRESCRIBINGID, PROCEDURESID, PROVIDERID, MEDADMIN_PROVIDERID, OBSGEN_PROVIDERID, OBSCLIN_PROVIDERID, RX_PROVIDERID, and VX_PROVIDERID) do not have harmonized field lengths	Data Model Conformance	Required	No	0
DC 1.16 ^c	Table IIF	Laboratory results or clinical observations are recorded in the wrong table based on the LOINC® classtype	Data Model Conformance	Investigative	No	11
DC 1.17	Table IID	Zip codes in the ENCOUNTER or LDS_ADDRESS_HISTORY table have non-conforming values (alphabetical characters or the wrong number of digits)	Data Model Conformance	Required	No	0
DC 1.18	Table ID	Table refresh dates are not documented in the HARVEST table for populated tables	Data Model Conformance	Required	No	0
DC 1.19	Table IIE	More than 10% of the values in any hash token field are assigned to multiple patients	Data Model Conformance	Investigative	No	4
DC 1.20	Table IIG	More than 5% of LOINC records in the LAB_RESULT_CM, PRO_CM, and OBS_CLIN tables are panel codes based on the LOINC® panel type	Data Model Conformance	Investigative	No	1.3
DC 1.21	Table IID	STATE_FIPS, COUNTY_FIPS and RUCA_ZIP fields have non-conforming values (alphabetical characters or discrepancies between STATE_FIPS and COUNTY_FIPS)	Data Model Conformance	Required	No	---
DC 2.01	Table IIIA	More than 5% of records have future dates. Future dates are defined as those with dates occurring after the maximum refresh date in the HARVEST table	Data Plausibility	Investigative	No	1.3
DC 2.02	Table IIIB	More than 10% of records fall into the lowest or highest categories of age (<0 or >89), height (<21 inches or >76 inches), weight (<0 lbs or >350 lbs), diastolic blood pressure (<40 mmHg or >120 mmHg), systolic blood pressure (<40 mmHg or >210 mmHg), or dispensed days supply (<1 day or >90 days)	Data Plausibility	Investigative	No	9
DC 2.03	Table IIIC	More than 5% of patients have illogical date relationships. Illogical date relationships are defined as dates of service which occur before a patient's birth date or after a patient's death date; procedure dates occurring more than 5 days before the admit date or 5 days after the discharge date for the same encounter; or stop dates before start dates in the EXTERNAL_MEDS, MED_ADMIN, OBS_CLIN, OBS_GEN, or PAT_RELATIONSHIP tables	Data Plausibility	Investigative	No	19

(a) Can be detected by CDM Diagnostic Query.

(b) Non-conforming values in most fields can be detected by the Value Set Conformance Query

(c) Can be detected by the Potential Code Errors Query

Data Check Exception Summary (continued - page 3 of 6)

This table summarizes conformance with PCORnet Data Checks v19. Changes to the previous cycle's data checks are explained in the footnotes. Exceptions to required data checks are highlighted in red and must be corrected before returning results. Exceptions to investigative data checks are highlighted in blue and must be investigated and explained in the ETL ADD. Network-wide results for PCORnet DataMarts approved as of November 3, 2025 are presented for comparison; a result of 10.3 means that 10.3% of DataMarts had exceptions to this data check.

Data Check	EDC Table	Data Check Description	Category	Type	Exception?	% DataMarts w/ exceptions
DC 2.04	Table IIID	The average number of encounters per visit is > 2.0 for inpatient (IP), emergency department (ED), or ED to inpatient (EI) encounters. A visit is defined as a unique combination of PATID, ENC_TYPE, ADMIT_DATE, and PROVIDERID in the ENCOUNTER table	Data Plausibility	Investigative	No	0
DC 2.07	Table IVE	The average number of principal diagnoses per known DX_ORIGIN per encounter is above threshold [2.0 for inpatient (IP) and ED to inpatient (EI)]	Data Plausibility	Investigative	No	23
DC 2.08	Table IIIG	The monthly volume of encounter, diagnosis, procedure, vital, prescribing, medication administration, external medications, laboratory records, or patient-reported outcomes is an outlier. Outliers are defined as months with 0 records or a significant decrease compared to the average volume in the previous 12 months. Encounters, diagnoses, and procedures are limited to ambulatory (AV), telehealth (TH), emergency department (ED), ED to inpatient (EI), or inpatient (IP) settings.	Data Plausibility	Investigative	No	28
DC 2.09	Table IB	Less than 80% of patients with a face-to-face encounter during the past 5 years have at least 1 face-to-face diagnosis and 1 vital measurement. Face-to-face is defined as an encounter type of ambulatory visit (AV), emergency department (ED), emergency department admit to inpatient hospital stay (EI), inpatient hospital (IP), or observation stay (OS).	Data Plausibility	Investigative	No	23
DC 3.01	Table IVA	The average number of diagnoses records with known diagnosis types per encounter is below threshold [1.0 for ambulatory (AV), inpatient (IP), emergency department (ED), ED to inpatient (EI), or telehealth (TH) encounters].	Data Completeness	Investigative	No	6.4
DC 3.02	Table IVB	The average number of procedure records with known procedure types per encounter is below threshold [0.75 for ambulatory (AV) encounters, 0.75 for emergency department (ED) encounters, 1.00 for ED to inpatient (EI) encounters, and 1.00 for inpatient (IP) encounters].	Data Completeness	Investigative	No	15

(a) Can be detected by CDM Diagnostic Query.

(b) Non-conforming values in most fields can be detected by the Value Set Conformance Query

(c) Can be detected by the Potential Code Errors Query

Data Check Exception Summary (continued - page 4 of 6)

This table summarizes conformance with PCORnet Data Checks v19. Changes to the previous cycle's data checks are explained in the footnotes. Exceptions to required data checks are highlighted in red and must be corrected before returning results. Exceptions to investigative data checks are highlighted in blue and must be investigated and explained in the ETL ADD. Network-wide results for PCORnet DataMarts approved as of November 3, 2025 are presented for comparison; a result of 10.3 means that 10.3% of DataMarts had exceptions to this data check.

Data Check	EDC Table	Data Check Description	Category	Type	Exception?	% DataMarts w/ exceptions
DC 3.03	Table IVC	More than 10% of records have missing or unknown values for the following fields: DISCHARGE_DISPOSITION (IP/EI encounters only), DISPENSE_SUP, DX_SOURCE, SEX, code fields [DEATH_CAUSE_CODE, MEDADMIN_CODE, OBSCLIN_CODE, OBSGEN_CODE], code type fields [DX_TYPE, CONDITION_TYPE, MEDADMIN_TYPE, OBSCLIN_TYPE, OBSGEN_TYPE, PX_TYPE, VX_CODE_TYPE], date fields [BIRTH_DATE, DISCHARGE_DATE (IP/EI encounters only), EXT_RECORD_DATE, RX_ORDER_DATE, PX_DATE, VX_RECORD_DATE], ENCOUNTERID fields in selected tables [DIAGNOSIS, LAB_RESULT_CM, PRESCRIBING, MED_ADMIN, OBS_CLIN, PROCEDURES, and VITAL], and provenance fields [CONDITION_SOURCE, DEATH_CAUSE_SOURCE, DEATH_SOURCE, DISPENSE_SOURCE, DX_ORIGIN, EXTMED_SOURCE, EXT_BASIS, MEDADMIN_SOURCE, LAB_RESULT_SOURCE, PX_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE]	Data Completeness	Investigative	No	62
DC 3.03	Table IVD	See Table IVC	Data Completeness	Investigative	Yes	62
DC 3.04	Table IB	Less than 50% of patients with encounters have DIAGNOSIS records	Data Completeness	Required	No	0
DC 3.05	Table IB	Less than 50% of patients with encounters have PROCEDURES records	Data Completeness	Required	No	0
DC 3.06	Table IVE	More than 10% of IP (inpatient) or ED to inpatient (EI) encounters with any diagnosis from a known DX_ORIGIN don't have a principal diagnosis from that source	Data Completeness	Investigative	No	15
DC 3.07	Table IVF	Encounters, diagnoses, or procedures in an ambulatory (AV), telehealth (TH), emergency department (ED), ED to inpatient (EI), or inpatient (IP) setting are less than 75% complete two months prior to the current month. Data completeness is calculated by comparing actual volume to the average volume during the previous year	Data Completeness	Investigative	No	15
DC 3.08	Table IVH	Less than 80% of prescribing orders are mapped to a RXCUI which fully specifies the ingredient, strength and dose form (i.e. RXCUI codes that have a Term Type of SCD, SBD,BPCK, or GPCK)	Data Completeness	Investigative	No	9.3

(a) Can be detected by CDM Diagnostic Query.

(b) Non-conforming values in most fields can be detected by the Value Set Conformance Query

(c) Can be detected by the Potential Code Errors Query

Data Check Exception Summary (continued - page 5 of 6)

This table summarizes conformance with PCORnet Data Checks v19. Changes to the previous cycle's data checks are explained in the footnotes. Exceptions to required data checks are highlighted in red and must be corrected before returning results. Exceptions to investigative data checks are highlighted in blue and must be investigated and explained in the ETL ADD. Network-wide results for PCORnet DataMarts approved as of November 3, 2025 are presented for comparison; a result of 10.3 means that 10.3% of DataMarts had exceptions to this data check.

Data Check	EDC Table	Data Check Description	Category	Type	Exception?	% DataMarts w/ exceptions
DC 3.09	Table IVI	Less than 80% of laboratory results are mapped to LAB_LOINC and have either a quantitative result (RESULT_NUM is not null and RESULT_MODIFIER is not NI, UN, OT, or null) or a qualitative result (RESULT_QUAL is not NI, UN, OT, or null)	Data Completeness	Investigative	No	7.9
DC 3.10	Table IVI	Less than 80% of quantitative results for tests mapped to LAB_LOINC fully specify the normal range in the range and modifier fields	Data Completeness	Investigative	No	28
DC 3.11	Table IVG	Vital, prescribing, or laboratory records are less than 75% complete three months prior to the current month. Data completeness is calculated by comparing actual volume to the average volume during the previous year	Data Completeness	Investigative	No	13
DC 3.12	Table IVI	Less than 80% of quantitative results for tests mapped to LAB_LOINC fully specify the RESULT_UNIT (i.e. RESULT_NUM is not null and RESULT_UNIT is not NI, UN, OT, or null)	Data Completeness	Investigative	No	7.9
DC 3.14	Table IVJ	Medication administration, dispensing, or clinical observation records are less than 75% complete three months prior to the current month. Data completeness is calculated by comparing actual volume to the average volume during the previous year	Data Completeness	Investigative	No	35
DC 3.15	Table IVH	Less than 80% of medication administrations mapped to RXNORM are mapped to a RXCUI that fully specifies the ingredient, strength and dose form (i.e. RXCUI codes that have a Term Type of SCD, SBD,BPCK, or GPCK)	Data Completeness	Investigative	No	6.3
DC 3.16	Table IVI	Less than 80% of clinical observations are mapped to an OBSCLIN_CODE and have a quantitative result (OBSCLIN_RESULT_NUM is not null and OBSCLIN_RESULT_MODIFIER is not NI,UN,OT or null), qualitative result (OBSCLIN_RESULT_QUAL is not NI,UN,OT or null) or narrative result (OBSCLIN_RESULT_TEXT is not null)	Data Completeness	Investigative	No	4.2
DC 3.17	Table IVI	Less than 80% of quantitative results for tests mapped to OBSCLIN_CODE fully specify the RESULT_UNIT (i.e. OBSCLIN_RESULT_NUM is not null and OBSCLIN_RESULT_UNIT is not NI, UN, OT, or null)	Data Completeness	Investigative	No	7.3

(a) Can be detected by CDM Diagnostic Query.

(b) Non-conforming values in most fields can be detected by the Value Set Conformance Query

(c) Can be detected by the Potential Code Errors Query

Data Check Exception Summary (continued - page 6 of 6)

This table summarizes conformance with PCORnet Data Checks v19. Changes to the previous cycle's data checks are explained in the footnotes. Exceptions to required data checks are highlighted in red and must be corrected before returning results. Exceptions to investigative data checks are highlighted in blue and must be investigated and explained in the ETL ADD. Network-wide results for PCORnet DataMarts approved as of November 3, 2025 are presented for comparison; a result of 10.3 means that 10.3% of DataMarts had exceptions to this data check.

Data Check	EDC Table	Data Check Description	Category	Type	Exception?	% DataMarts w/ exceptions
DC 4.01	Table VA	More than a 5% decrease in the number of patients or records in a CDM table between the previous and current DataMart refresh	Data Persistence	Investigative	No	32
DC 4.02	Table VB	More than a 5% decrease in the number of patients or records for diagnosis, procedures, labs or prescriptions during an ambulatory (AV), telehealth (TH), other ambulatory (OA), emergency department (ED), or inpatient (IP) encounter between the previous and current DataMart refresh	Data Persistence	Investigative	No	27
DC 4.03	Table VC	More than a 5% decrease in the number of records or distinct codes for CPT/HCPCS, CVX, ICD10, LOINC, NDC, or RXNORM codes between the previous and current DataMart refresh	Data Persistence	Investigative	No	27

(a) Can be detected by CDM Diagnostic Query.

(b) Non-conforming values in most fields can be detected by the Value Set Conformance Query

(c) Can be detected by the Potential Code Errors Query

Table IA. Demographic Summary

This table contains general descriptive information about the patients in the DEMOGRAPHIC table. These patients may or may not be represented in other CDM tables.

	N	%	Source table
Patients	2,284,557		DEM_L3_N
Age			DEM_L3_AGEYRSDIST1
Mean	51		
Median	51		
Age group			DEM_L3_AGEYRSDIST2
0-4	36,077	1.6	
5-14	117,695	5.1	
15-21	128,107	5.6	
22-64	1,272,933	55.8	
65+	720,559	31.5	
Missing	9,186	0.4	
Hispanic			DEM_L3_HISPDIST
N (No)	1,138,403	49.8	
Y (Yes)	36,303	1.6	
Missing or Refused	1,109,851	48.6	
Sex			DEM_L3_SEXDIST
F (Female)	1,154,641	50.5	
M (Male)	1,112,438	48.7	
Missing or Ambiguous	17,478	0.8	
Race			DEM_L3_RACEDIST
White	1,365,611	59.8	
Non-White	145,068	6.4	
Missing or Refused	773,878	33.8	

The four 'flavors of null' defined in the CDM are combined in the missing categories shown here, but details are available in the source tables.

Age is calculated as current age or age at death if death date is known. If multiple death records exist, the earlier death date is used.

Percentages are sums of the percentages in the source table and are subject to rounding errors.

Race distribution is shown for all patients and for patients seen after Meaningful Use standards were widely implemented.

Table IA. Demographic Summary (continued)

This table contains general descriptive information about the patients in the DEMOGRAPHIC table. These patients may or may not be represented in other CDM tables.

	N	%	Source table
Race/Ethnicity			
American Indian or Alaska Native	0		DEM_L3_RACEETHAIAN
Asian	0		DEM_L3_RACEETHASIAN
Black	0		DEM_L3_RACEETHBLACK
Hispanic	0		DEM_L3_RACEETHHISPANIC
Middle Eastern or North African	0		DEM_L3_RACEETHMENA
Native Hawaiian or Pacific Islander	0		DEM_L3_RACEETHNHPI
White	0		DEM_L3_RACEETHWHITE
Missing or Refused	2,284,557	100.0	DEM_L3_RACEETHMISS
Gender Identity			DEM_L3_GENDERDIST
GQ (Genderqueer/Non-Binary)	0	0.0	
M (Man)	0	0.0	
W (Woman)	0	0.0	
MU (Multiple gender categories), SE (Something else), TF (Transgender female/Trans woman/Male-to-female), or TM (Transgender male/Trans man/Female-to-male)	0	0.0	
Missing or Refused	2,284,557	100.0	
Sexual Orientation			DEM_L3_ORIENTDIST
Bisexual	0	0.0	
Gay	0	0.0	
Lesbian	0	0.0	
Straight	0	0.0	
AS (Asexual), MU (Multiple sexual orientations), SE (Something else), QS (Questioning)	0	0.0	
Missing or Refused	2,284,557	100.0	
Queer	0	0.0	

The four 'flavors of null' defined in the CDM are combined in the missing categories shown here, but details are available in the source tables.

Age is calculated as current age or age at death if death date is known. If multiple death records exist, the earlier death date is used.

Percentages are sums of the percentages in the source table and are subject to rounding errors.

Race distribution is shown for all patients and for patients seen after Meaningful Use standards were widely implemented.

Table IB. Potential Pools of Patients

This table illustrates the number of patients meeting different inclusion criteria and supports Data Check 2.09 (Less than 80% of patients with a face-to-face encounter during the past 5 years have at least 1 face-to-face diagnosis and 1 vital measurement), Data Check 3.04 (less than 50% of patients with encounters have DIAGNOSIS records) and Data Check 3.05 (less than 50% of patients with encounters have PROCEDURES records). Data check exceptions to 3.04 and 3.05 are highlighted in red and must be corrected; data check exceptions to 2.09 are highlighted in blue and must be explained in the ETL ADD.

Metric	Metric Description	Result	%	Source table
All patients	Number of unique patients in the DEMOGRAPHIC table	2,284,557		DEM_L3_N
Potential pool of patients for observational studies	Number of unique patients with at least 1 face-to-face (ED, EI, IP, OS, or AV) encounter within the past 5 years	612,087	27	ENC_L3_DASH2
Potential pool of patients for trials	Number of unique patients with at least 1 face-to-face (ED, EI, IP, OS, or AV) encounter within the past 1 year	320,155	14	ENC_L3_DASH2
Potential pool of patients for studies requiring data on diagnoses, vital measures and (a) medications or (b) medications and lab results	Number of unique patients with at least 1 DIAGNOSIS record in a face-to-face setting and at least 1 VITAL record within the past 5 years	547,468	89.4	XTBL_L3_DASH1
	Number of unique patients with at least 1 DIAGNOSIS record in a face-to-face setting, at least 1 VITAL record, and at least 1 PRESCRIBING or MED_ADMIN record within the past 5 years	483,553	79.0	XTBL_L3_DASH2
	Number of unique patients with at least 1 DIAGNOSIS record in a face-to-face setting, at least 1 VITAL record, at least 1 PRESCRIBING or MED_ADMIN record, and at least 1 LAB_RESULT_CM record within the past 5 years	361,363	59.0	XTBL_L3_DASH3
Patients with diagnosis data	Percentage of patients with encounters who have at least 1 diagnosis	98%		ENC_L3_N; DIA_L3_N
Patients with procedure data	Percentage of patients with encounters who have at least 1 procedure	97%		ENC_L3_N; PRO_L3_N

Table IC. Height, Weight, and Body Mass Index

This table contains descriptive statistics and frequencies of VITAL measurements.

	Result	%	Source table
Height measurements			VIT_L3_HT_DIST
Records	7,510,913		
Height (inches), mean	64		
Height (inches), median	66		
Weight measurements			VIT_L3_WT_DIST
Records	10,227,034		
Weight (lbs.), mean	170		
Weight (lbs.), median	173		
Body Mass Index (BMI) measurements			VIT_L3_BMI
Records	5,789,455		
BMI <=25	2,103,707	36.3	
BMI 26-30	1,380,651	23.8	
BMI >=31	2,305,097	39.8	

Table ID. Records, Refresh Dates, Patients, Encounters, and Date Ranges by Table

This table displays record counts and refresh dates for all populated tables, and summary counts and date ranges for the characterized data. The characterized data may be a subset of all data for all tables except for DEMOGRAPHIC, DEATH_CAUSE, HASH_TOKEN, LAB_HISTORY, LDS_ADDRESS_HISTORY, and PROVIDER. Differences are due to the lookback date restriction and/or the exclusion of LOINC codes that may contain PII from the OBS_CLIN, OBS_GEN, LAB_RESULT_CM and PRO_CM tables. These data support Data Check 1.18 (table refresh dates are not documented). Data check exceptions are highlighted in red and must be corrected.

Table	All Data		Characterized Data			Characterized Data Range			Source Tables
	Records	Refresh Date	Records	Patients	Encounters	Field name	5th Percentile	95th Percentile	
DEMOGRAPHIC	2,284,557	05APR2026	2,284,557	2,284,557	---	BIRTH_DATE	1921_Oct	2014_Feb	DEM_L3_N; XTBL_L3_DATES
ENROLLMENT	1,352,491	05APR2026	830,128	830,128	---	ENR_START_DATE^	2000_Feb	2025_Apr	ENR_L3_N; XTBL_L3_DATES
ENCOUNTER	30,727,252	05APR2026	15,658,577	829,623	15,658,577	ADMIT_DATE	2016_Dec	2025_Nov	ENC_L3_N; XTBL_L3_DATES
DIAGNOSIS	83,746,587	03APR2026	53,705,073	812,736	13,699,321	ADMIT_DATE	2016_Dec	2025_Dec	DIA_L3_N; XTBL_L3_DATES
PROCEDURES	107,556,506	04APR2026	58,774,019	808,780	13,463,419	ADMIT_DATE	2016_Nov	2025_Dec	PRO_L3_N; XTBL_L3_DATES
DEATH	454,934	05APR2026	454,934	373,556	---	DEATH_DATE	1988_May	2024_May	DEATH_L3_N; XTBL_L3_DATES
VITAL	54,389,064	31MAR2026	36,127,241	861,005	8,874,700	MEASURE_DATE	2016_Dec	2025_Oct	VIT_L3_N; XTBL_L3_DATES
LAB_RESULT_CM	231,727,498	31MAR2026	167,597,234	541,068	3,362,834	RESULT_DATE	2017_Jan	2025_Nov	LAB_L3_N; XTBL_L3_DATES
PRESCRIBING	112,647,116	05APR2026	79,571,202	760,185	6,026,515	RX_ORDER_DATE	2016_Dec	2025_Nov	PRES_L3_N; XTBL_L3_DATES

^The lookback date for ENROLLMENT considers either ENR_START_DATE or ENR_END_DATE, but only ENR_START_DATE is displayed here. Descriptive statistics for ENR_END_DATE are available in the XTBL_L3_DATES dataset.

In the All Data columns, records comes from the DATAMART_ALL table and Refresh Date comes from the XTBL_L3_REFRESH_DATE table.

Records are the sum of ALL_N and NULL_N for the field which is the primary key for the table. Patients are the sum of DISTINCT_N and NULL_N for the PATID field. Encounters are the sum of DISTINCT_N and NULL_N for the ENCOUNTERID field.

Table ID. Records, Refresh Dates, Patients, Encounters, and Date Ranges by Table (continued - page 2 of 3)

This table displays record counts and refresh dates for all populated tables, and summary counts and date ranges for the characterized data. The characterized data may be a subset of all data for all tables except for DEMOGRAPHIC, DEATH_CAUSE, HASH_TOKEN, LAB_HISTORY, LDS_ADDRESS_HISTORY, and PROVIDER. Differences are due to the lookback date restriction and/or the exclusion of LOINC codes that may contain PII from the OBS_CLIN, OBS_GEN, LAB_RESULT_CM and PRO_CM tables. These data support Data Check 1.18 (table refresh dates are not documented). Data check exceptions are highlighted in red and must be corrected.

Table	All Data		Characterized Data			Characterized Data Range			Source Tables
	Records	Refresh Date	Records	Patients	Encounters	Field name	5th Percentile	95th Percentile	
DISPENSING	5,280,863	06APR2026	4,824,060	252,357	---	DISPENSE_DATE	2016_Nov	2025_Nov	DISP_L3_N; XTBL_L3_DATES
CONDITION	3,181,762	31MAR2026	2,791,908	477,354	0	REPORT_DATE	2017_Jun	2026_Jan	COND_L3_N; XTBL_L3_DATES
DEATH_CAUSE	19,017	05APR2026	19,017	18,601	---	---	---	---	DEATHC_L3_N
PRO_CM	42,983,172	31MAR2026	29,359,922	756,435	6,521,417	PRO_DATE	2016_Nov	2025_Nov	PROCM_L3_N; XTBL_L3_DATES
PROVIDER	38,590	05APR2026	38,590	---	---	---	---	---	PROV_L3_N
MED_ADMIN	53,526,068	31MAR2026	38,479,890	513,380	2,353,446	MEDADMIN_ START_DATE	2016_Nov	2025_Nov	MEDADM_L3_N; XTBL_L3_DATES
OBS_CLIN	744,764,460	01APR2026	486,423,175	862,879	9,128,855	OBSCLIN_START_ DATE	2016_Dec	2025_Nov	OBSCLIN_L3_N; XTBL_L3_DATES
OBS_GEN	91,749,641	03APR2026	87,218,390	837,440	6,842,057	OBSGEN_START_ DATE	2020_Mar	2025_Dec	OBSGEN_L3_N; XTBL_L3_DATES
PCORNET_TRIAL	356	05APR2026	356	355	---	---	---	---	TRIAL_L3_N

^The lookback date for ENROLLMENT considers either ENR_START_DATE or ENR_END_DATE, but only ENR_START_DATE is displayed here. Descriptive statistics for ENR_END_DATE are available in the XTBL_L3_DATES dataset.

In the All Data columns, records comes from the DATAMART_ALL table and Refresh Date comes from the XTBL_L3_REFRESH_DATE table.

Records are the sum of ALL_N and NULL_N for the field which is the primary key for the table. Patients are the sum of DISTINCT_N and NULL_N for the PATID field. Encounters are the sum of DISTINCT_N and NULL_N for the ENCOUNTERID field.

Table ID. Records, Refresh Dates, Patients, Encounters, and Date Ranges by Table (continued - page 3 of 3)

This table displays record counts and refresh dates for all populated tables, and summary counts and date ranges for the characterized data. The characterized data may be a subset of all data for all tables except for DEMOGRAPHIC, DEATH_CAUSE, HASH_TOKEN, LAB_HISTORY, LDS_ADDRESS_HISTORY, and PROVIDER. Differences are due to the lookback date restriction and/or the exclusion of LOINC codes that may contain PII from the OBS_CLIN, OBS_GEN, LAB_RESULT_CM and PRO_CM tables. These data support Data Check 1.18 (table refresh dates are not documented). Data check exceptions are highlighted in red and must be corrected.

Table	All Data		Characterized Data			Characterized Data Range			Source Tables
	Records	Refresh Date	Records	Patients	Encounters	Field name	5th Percentile	95th Percentile	
HASH_TOKEN	2,284,557	05APR2026	2,284,557	2,284,557	---	---	---	---	HASH_L3_N
LDS_ADDRESS_HISTORY	2,852,571	05APR2026	2,852,571	2,207,036	---	---	---	---	LDSADRS_L3_N
IMMUNIZATION	5,035,750	31MAR2026	2,436,968	375,540	921,713	VX_RECORD_DATE	2016_Nov	2025_Mar	IMMUNE_L3_N; XTBL_L3_DATES
LAB_HISTORY	17,694	---	17,694	---	---	---	---	---	LABHIST_L3_N
EXTERNAL_MEDS	8,031,171	N/A	4,599,717	551,286	---	EXT_RECORD_DATE	2016_Nov	2025_Sep	EXTMED_L3_N; XTBL_L3_DATES
PAT_RELATIONSHIP	82,582	N/A	82,582	55,543	---	RELATIONSHIP_START			PATREL_L3_N; XTBL_L3_DATES

^The lookback date for ENROLLMENT considers either ENR_START_DATE or ENR_END_DATE, but only ENR_START_DATE is displayed here. Descriptive statistics for ENR_END_DATE are available in the XTBL_L3_DATES dataset.

In the All Data columns, records comes from the DATAMART_ALL table and Refresh Date comes from the XTBL_L3_REFRESH_DATE table.

Records are the sum of ALL_N and NULL_N for the field which is the primary key for the table. Patients are the sum of DISTINCT_N and NULL_N for the PATID field. Encounters are the sum of DISTINCT_N and NULL_N for the ENCOUNTERID field.

Table IE. Records Per Table By Encounter Type

This table contains record counts by encounter type for the ENCOUNTER, DIAGNOSIS, and PROCEDURES tables.

Encounter Type	ENCOUNTER		DIAGNOSIS		PROCEDURES	
	N	%	N	%	N	%
AV (Ambulatory Visit)	14,054,751	89.8	39,314,798	73.2	39,945,453	68.0
ED (Emergency Dept)	830,269	5.3	4,348,678	8.1	5,076,370	8.6
EI (ED to IP Stay)	176,236	1.1	5,546,492	10.3	7,719,595	13.1
IC (Institutional Professional Consult)	0		0		0	
IP (Inpatient Hospital Stay)	151,292	1.0	3,209,018	6.0	5,436,994	9.3
IS (Non-acute Institutional Stay)	0		0		0	
OA (Other Ambulatory Visit)	0		0		0	
OS (Observation Stay)	0		0		0	
TH (Telehealth)	446,029	2.8	1,286,087	2.4	595,607	1.0
Missing, NI, UN or OT	0		0		0	
Total	15,658,577		53,705,073		58,774,019	
Source table	ENC_L3_ENCTYPE		DIA_L3_ENCTYPE		PRO_L3_ENCTTPE	

Table IF. Date Obfuscation or Imputation

This table contains information about the presence of date obfuscation or imputation. Imputed or obfuscated dates are important to consider when interpreting results. The use of date obfuscation is highlighted in orange.

Table	Field	DATE_MGMT	Source table
HARVEST	BIRTH_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	ENR_START_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	ENR_END_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	ADMIT_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	DISCHARGE_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	PX_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	RX_ORDER_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	RX_START_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	RX_END_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	DISPENSE_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	LAB_ORDER_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	SPECIMEN_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	RESULT_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	MEASURE_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	ONSET_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	REPORT_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	RESOLVE_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	PRO_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	DEATH_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	MEDADMIN_START_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	MEDADMIN_STOP_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	OBSCLIN_START_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	OBSCLIN_STOP_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	OBSGEN_START_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	OBSGEN_STOP_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	DX_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	ADDRESS_PERIOD_START_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	ADDRESS_PERIOD_END_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	VX_RECORD_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	VX_ADMIN_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA
HARVEST	VX_EXP_DATE_MGMT	01 (No imputation or obfuscation)	XTBL_L3_METADATA

Details regarding the type of obfuscation or imputation are available in the source data.

Chart IA. Trend In Vital Measures by Measurement Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the VITAL table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

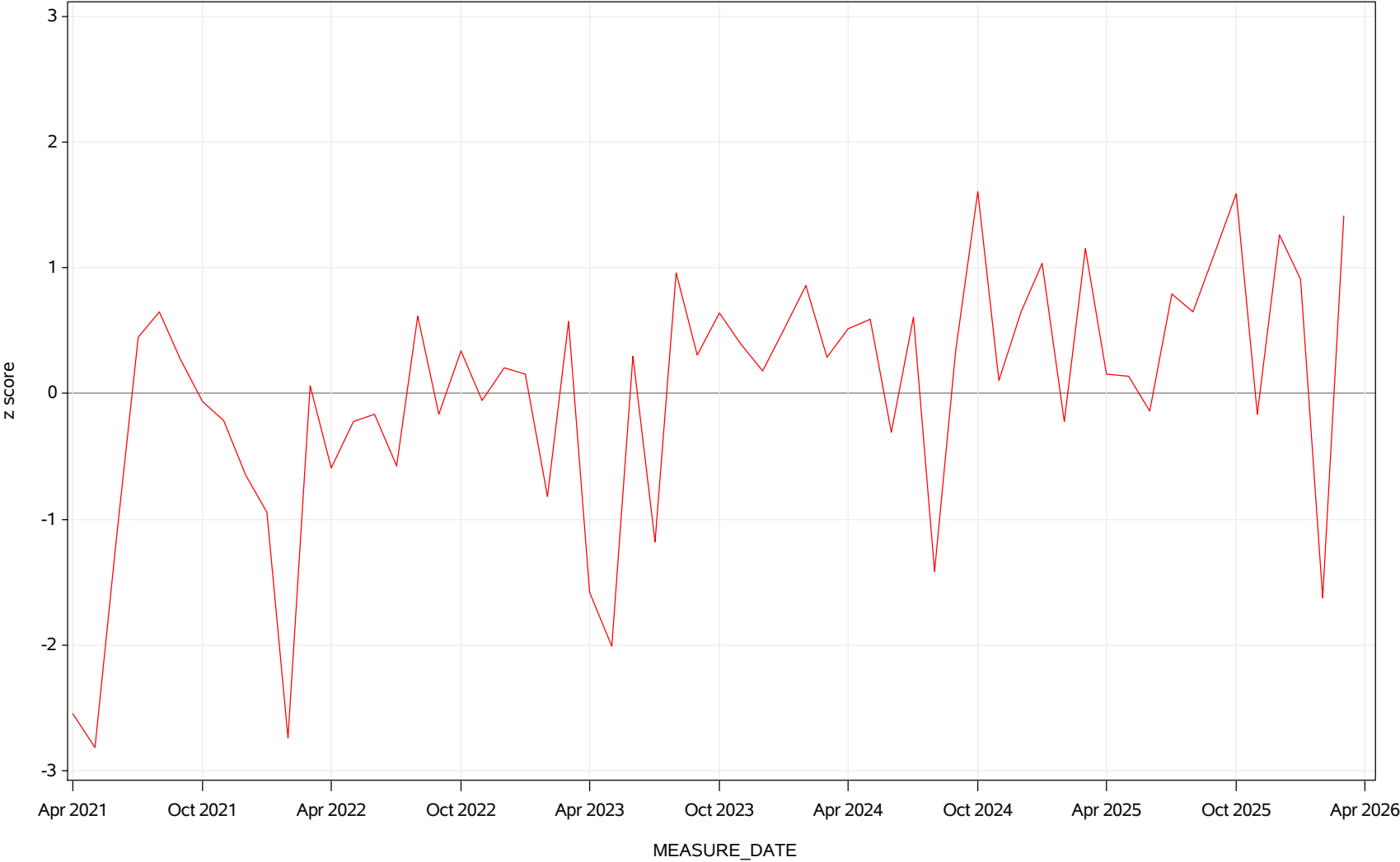


Chart IB. Trend in Encounters by Admit Date and Encounter Type, Past 5 Years

This 3 panel chart illustrates relative changes over time in the number of records per encounter type found in the ENCOUNTER table by admit date. The first panel includes AV, OA, and TH encounter types; the second panel includes ED, EI and IP encounter types, and the third panel includes IS, IC, and OS encounter types. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

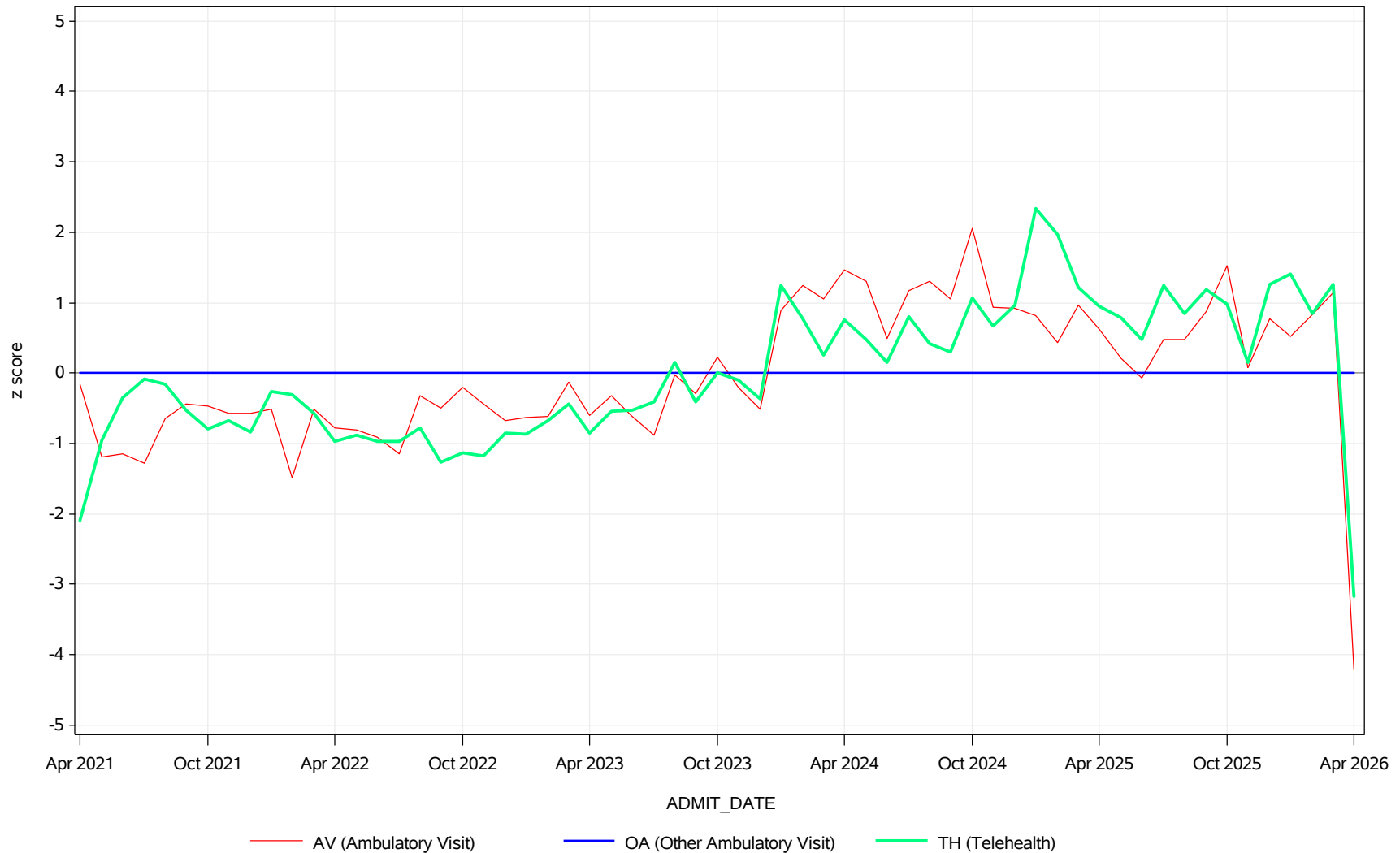


Chart IB. Trend in Encounters by Admit Date and Encounter Type, Past 5 Years (continued)

This 3 panel chart illustrates relative changes over time in the number of records per encounter type found in the ENCOUNTER table by admit date. The first panel includes AV, OA, and TH encounter types; the second panel includes ED, EI and IP encounter types, and the third panel includes IS, IC, and OS encounter types. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

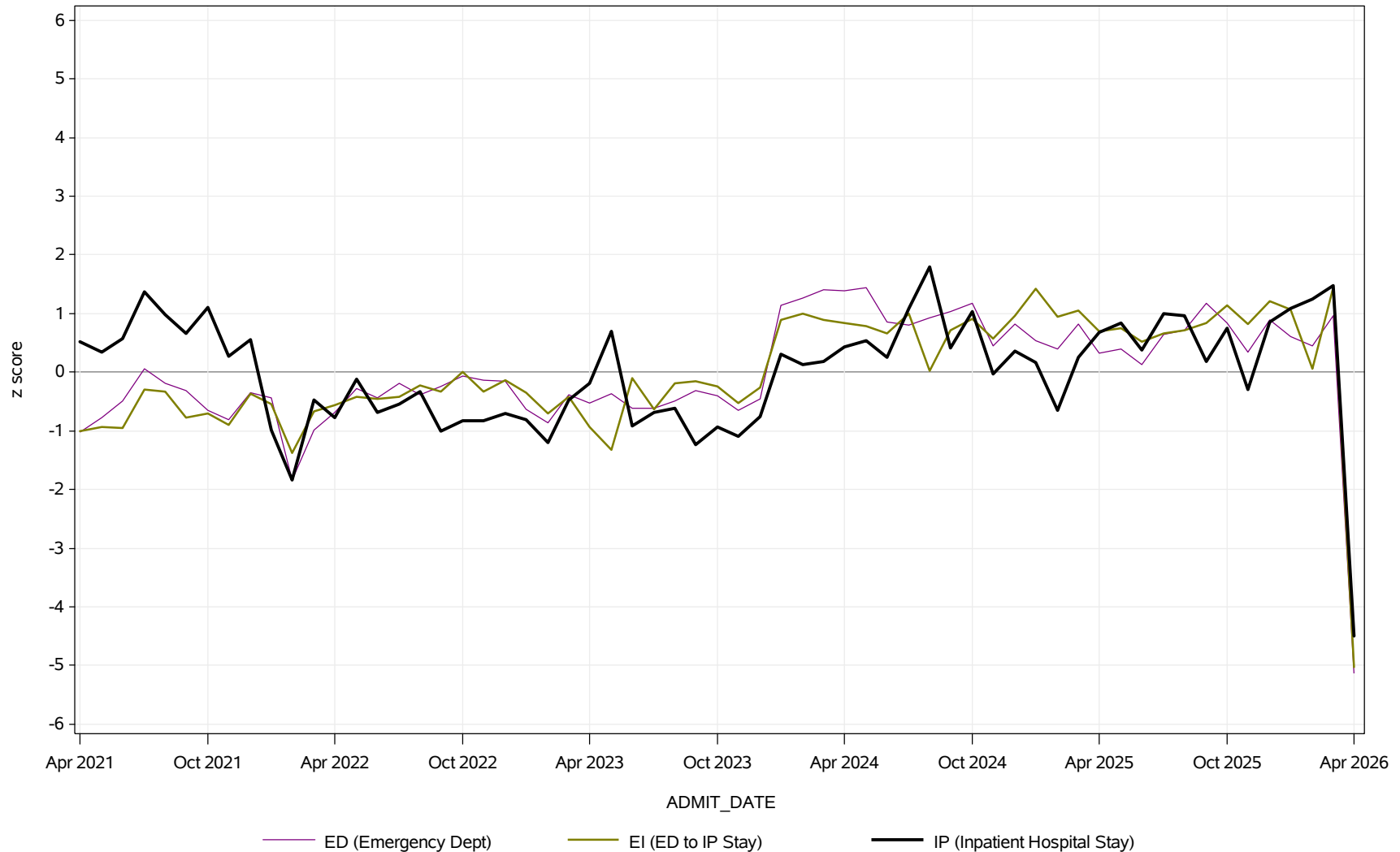


Chart IC. Trend in Institutional Encounters by Discharge Date and Encounter Type, Past 5 Years

This chart illustrates relative changes over time in the number of records for IP, IS, and EI encounter types found in the ENCOUNTER table by discharge date. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

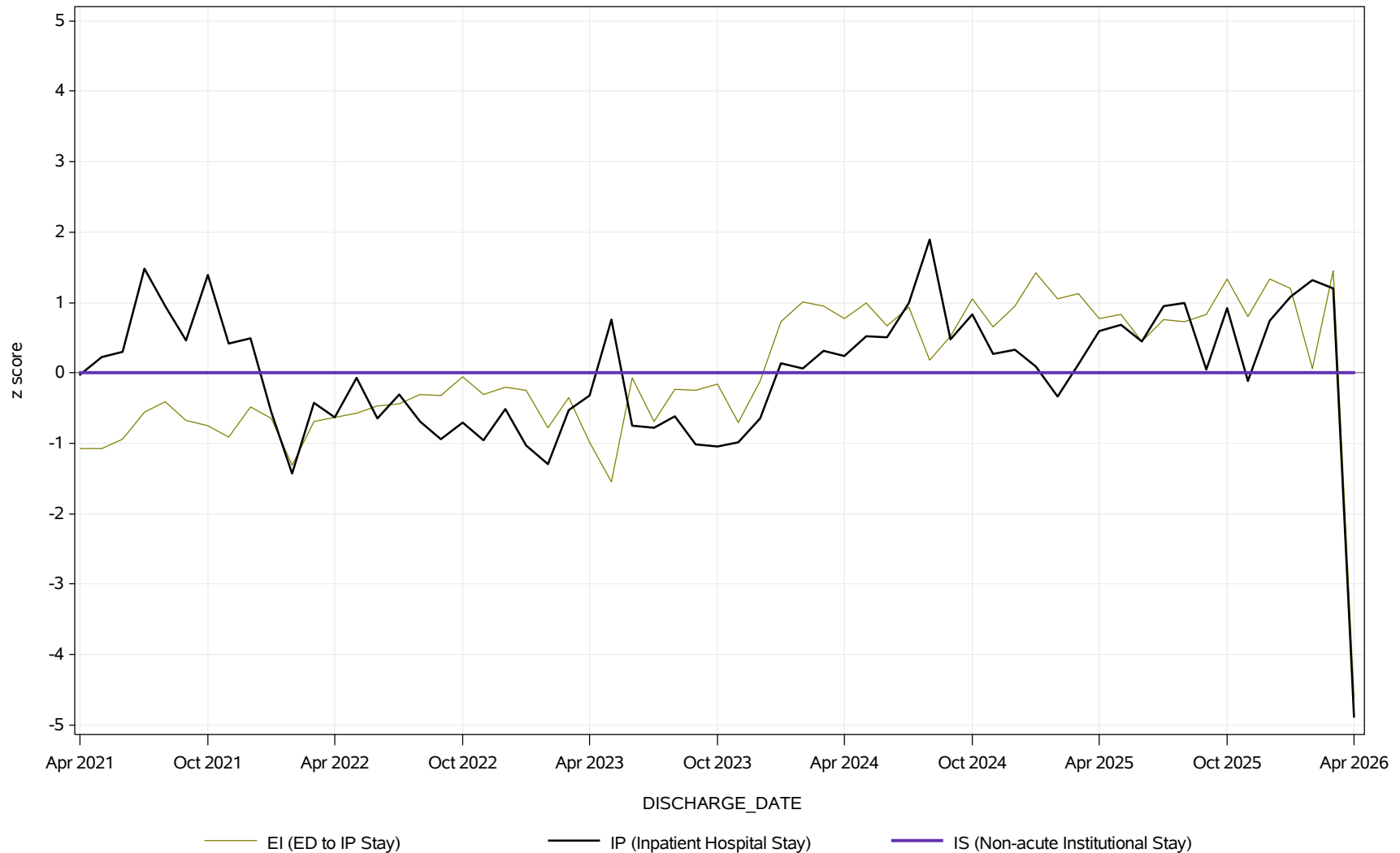


Chart ID. Trend in Lab Results By Result Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the LAB_RESULT_CM table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

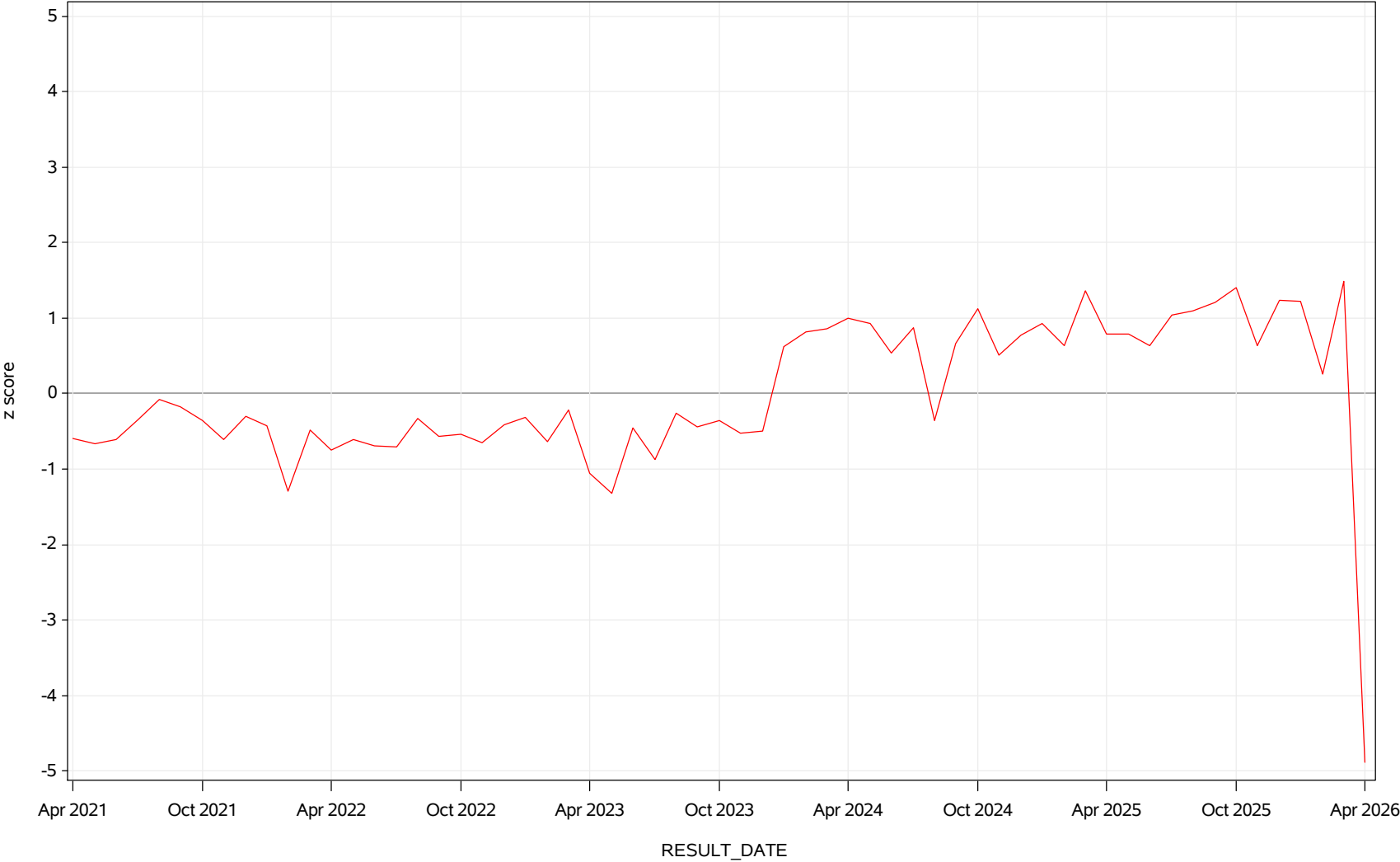


Chart IE. Trend in Prescribed Medications by Rx Order Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the PRESCRIBING table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated. Significant inflection points and other unexpected patterns should be investigated.

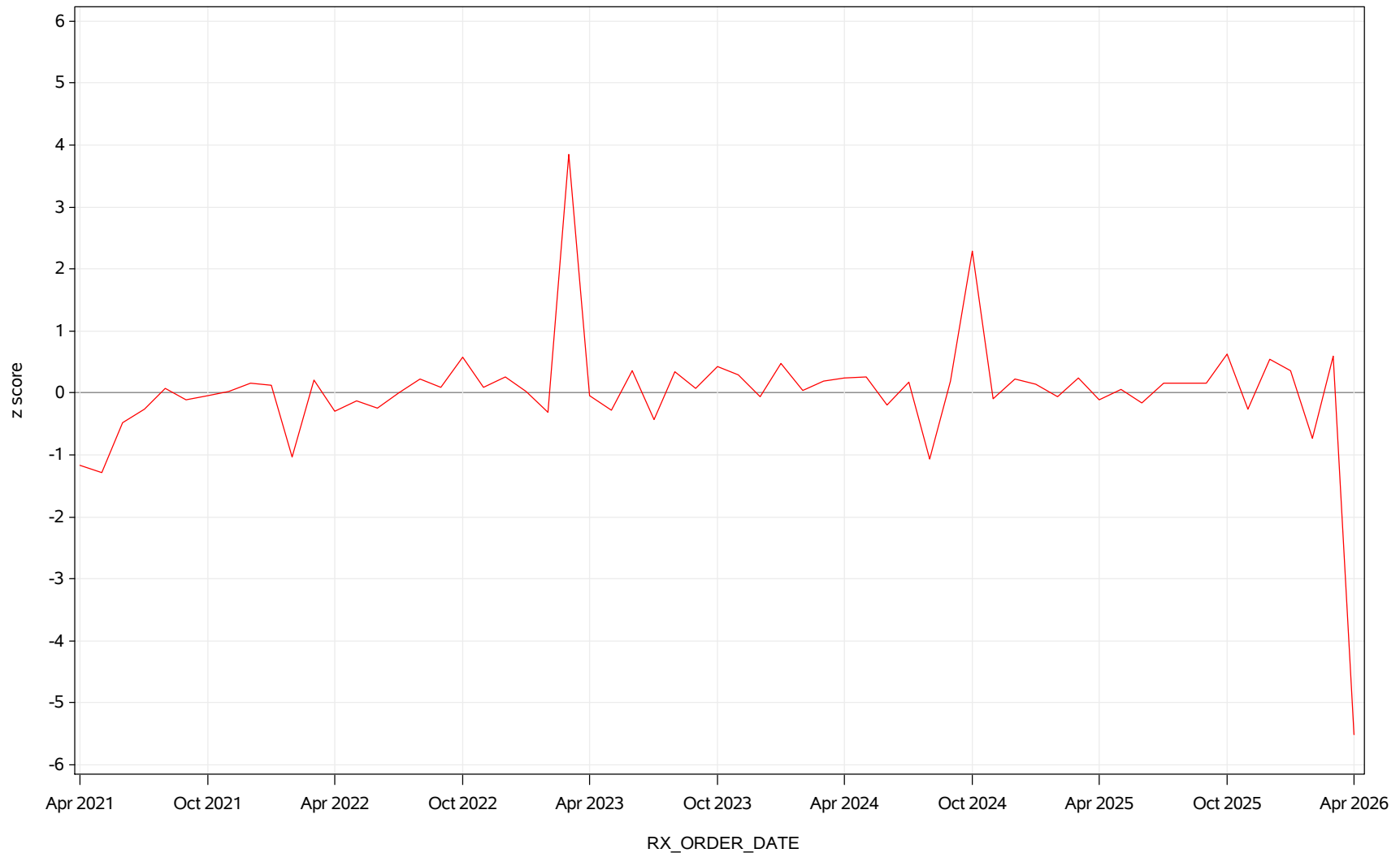


Chart IF. Trend in Dispensed Medications by Dispense Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the DISPENSING table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

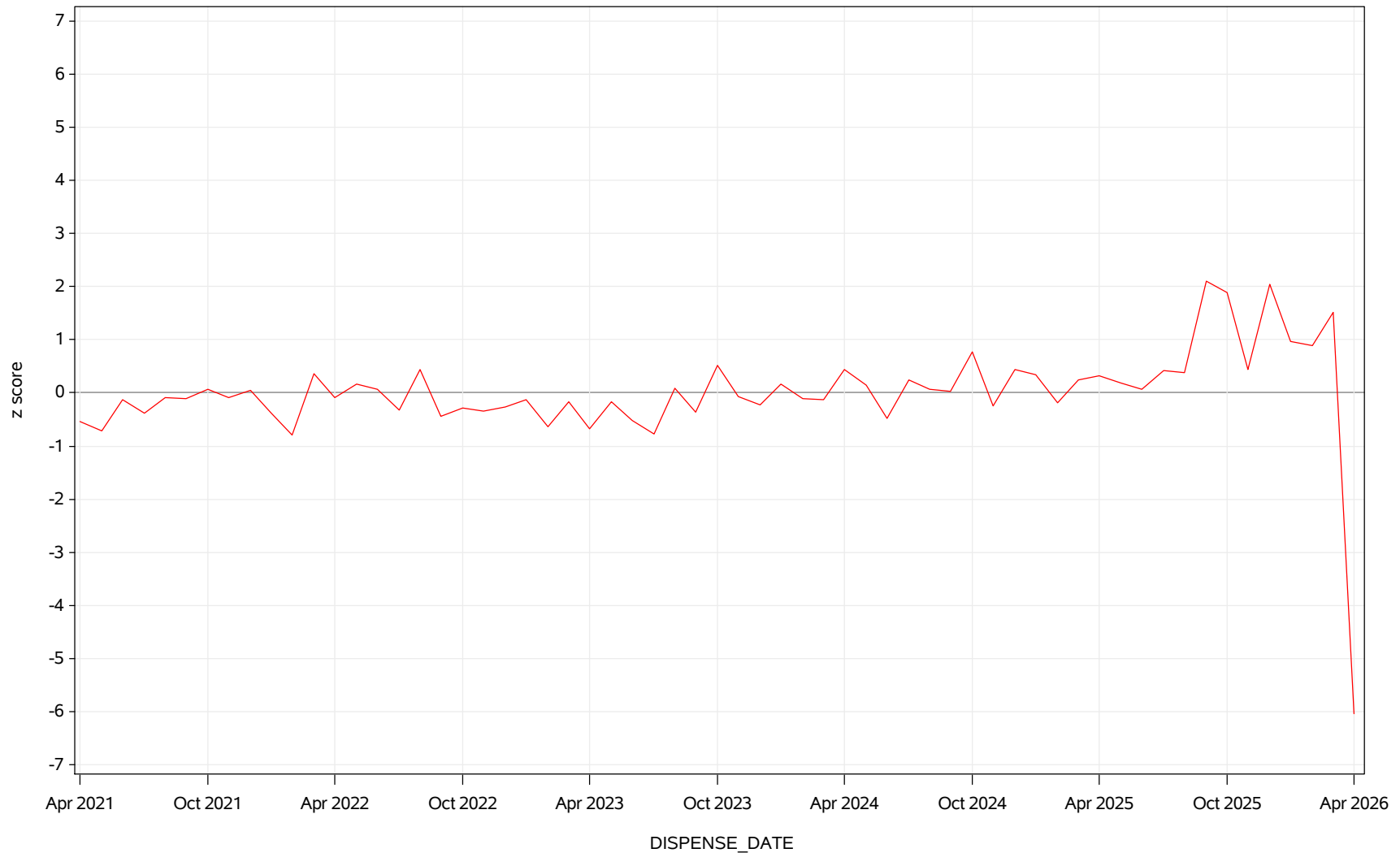


Chart IG. Trend in Administered Medications by Start Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the MED_ADMIN table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

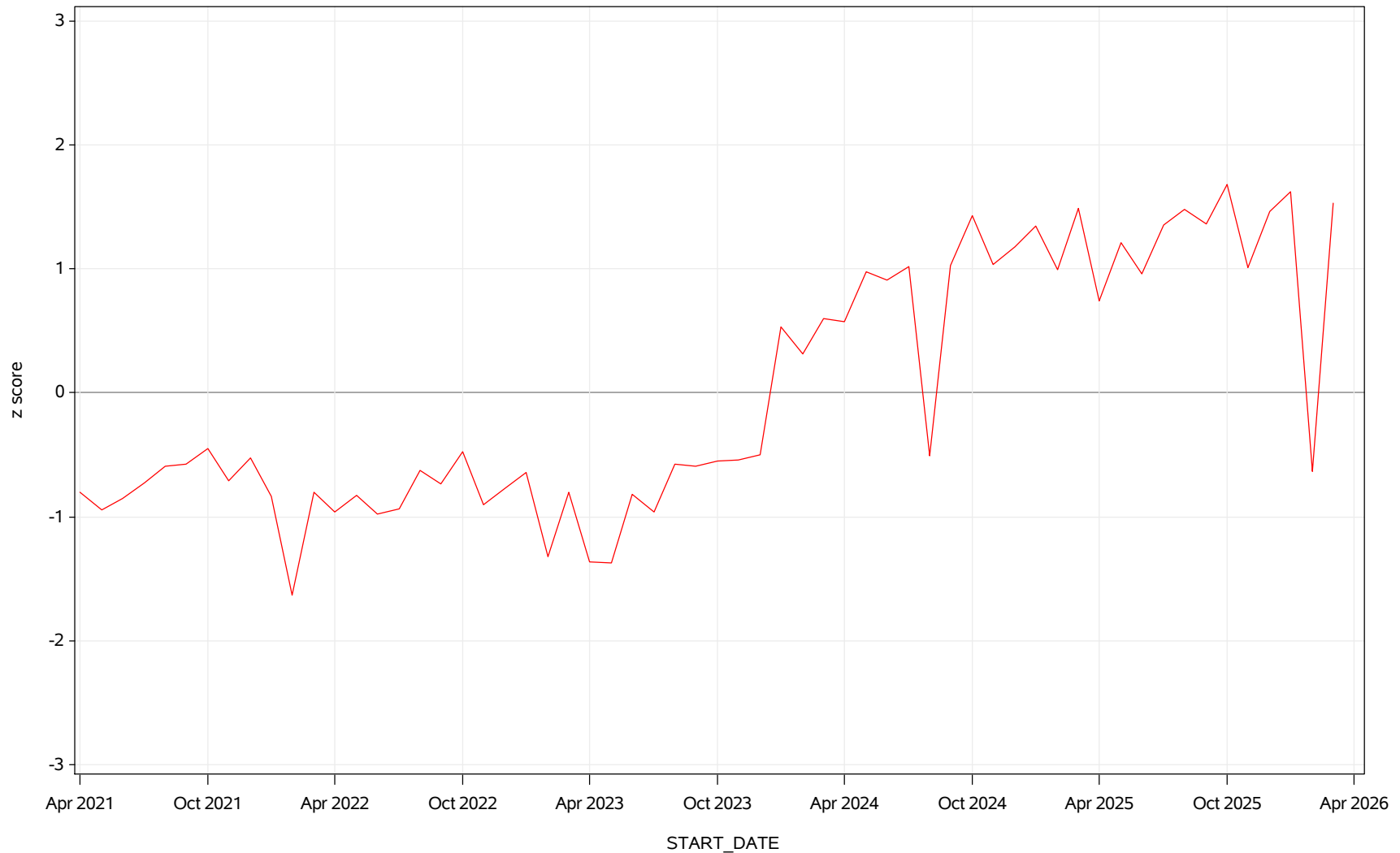


Chart IH. Trend in Condition Records by Report Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the CONDITION table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

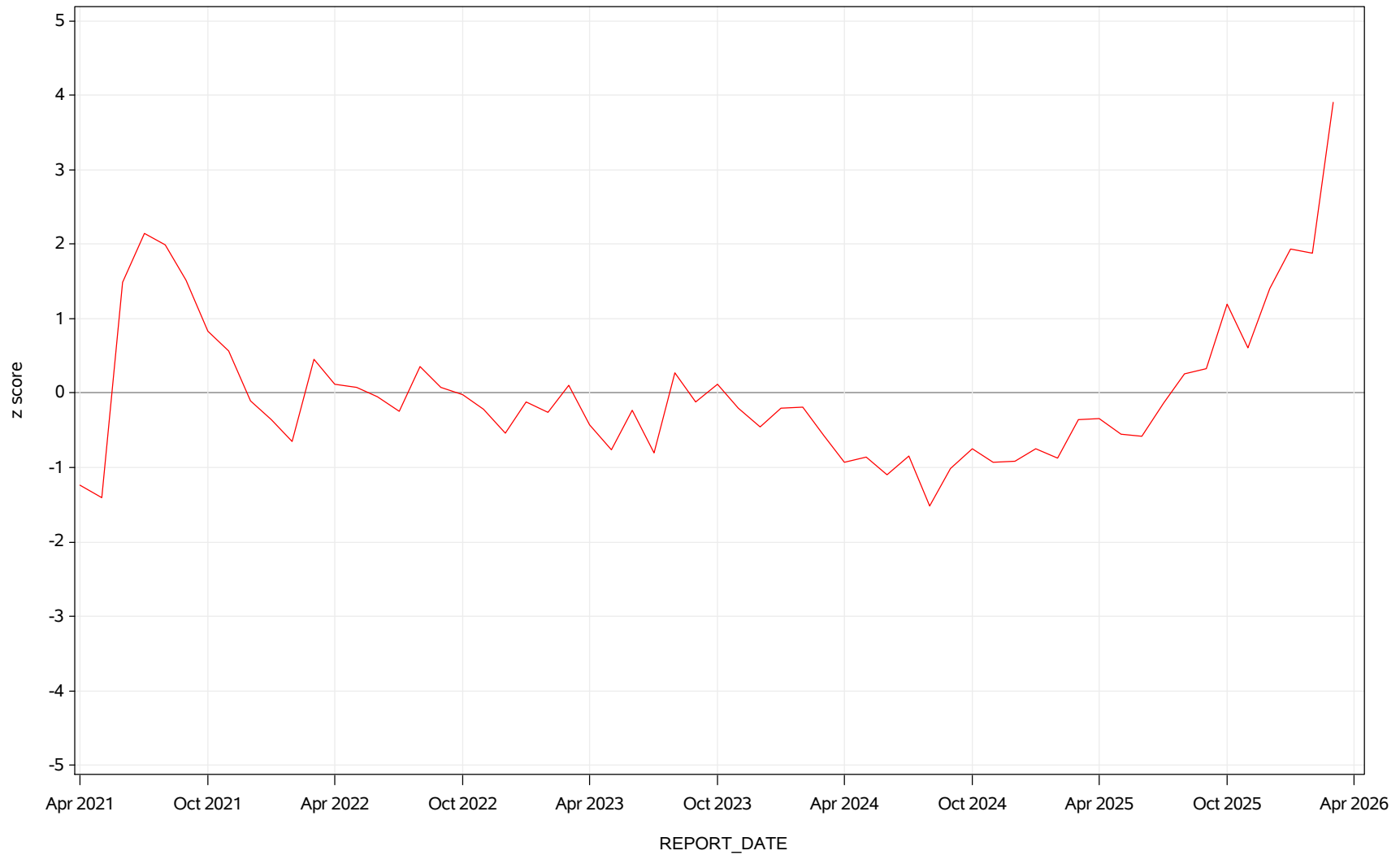


Chart II. Trend in Death Records by Death Date and Source, Past 5 Years

This chart illustrates relative changes over time in the number of records per death source found in the DEATH table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

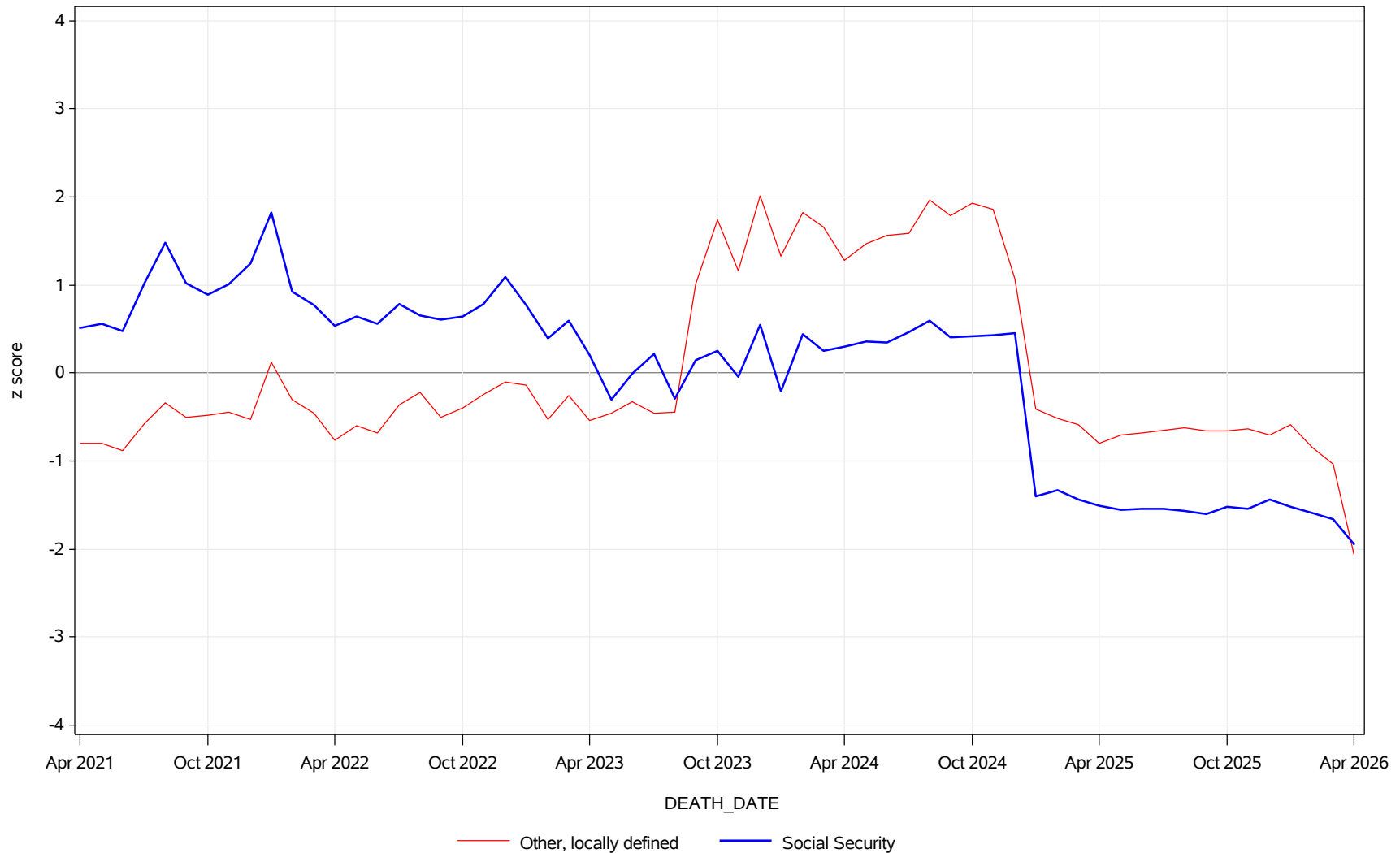


Chart IJ. Trend in Immunization Records by Vx Record Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the IMMUNIZATION table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

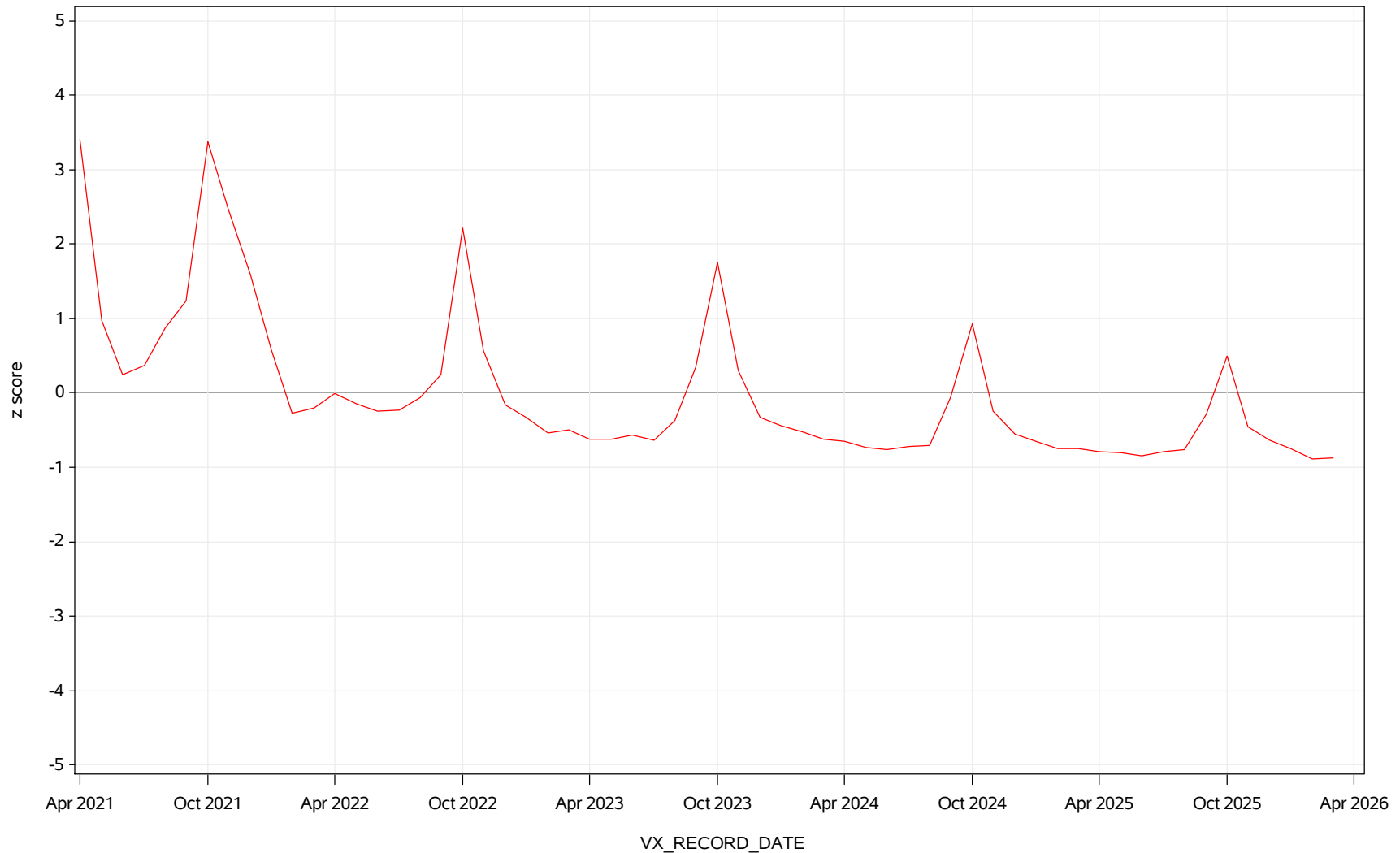


Chart IK. Trend in Clinical Observation Records by Start Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the OBS_CLIN table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

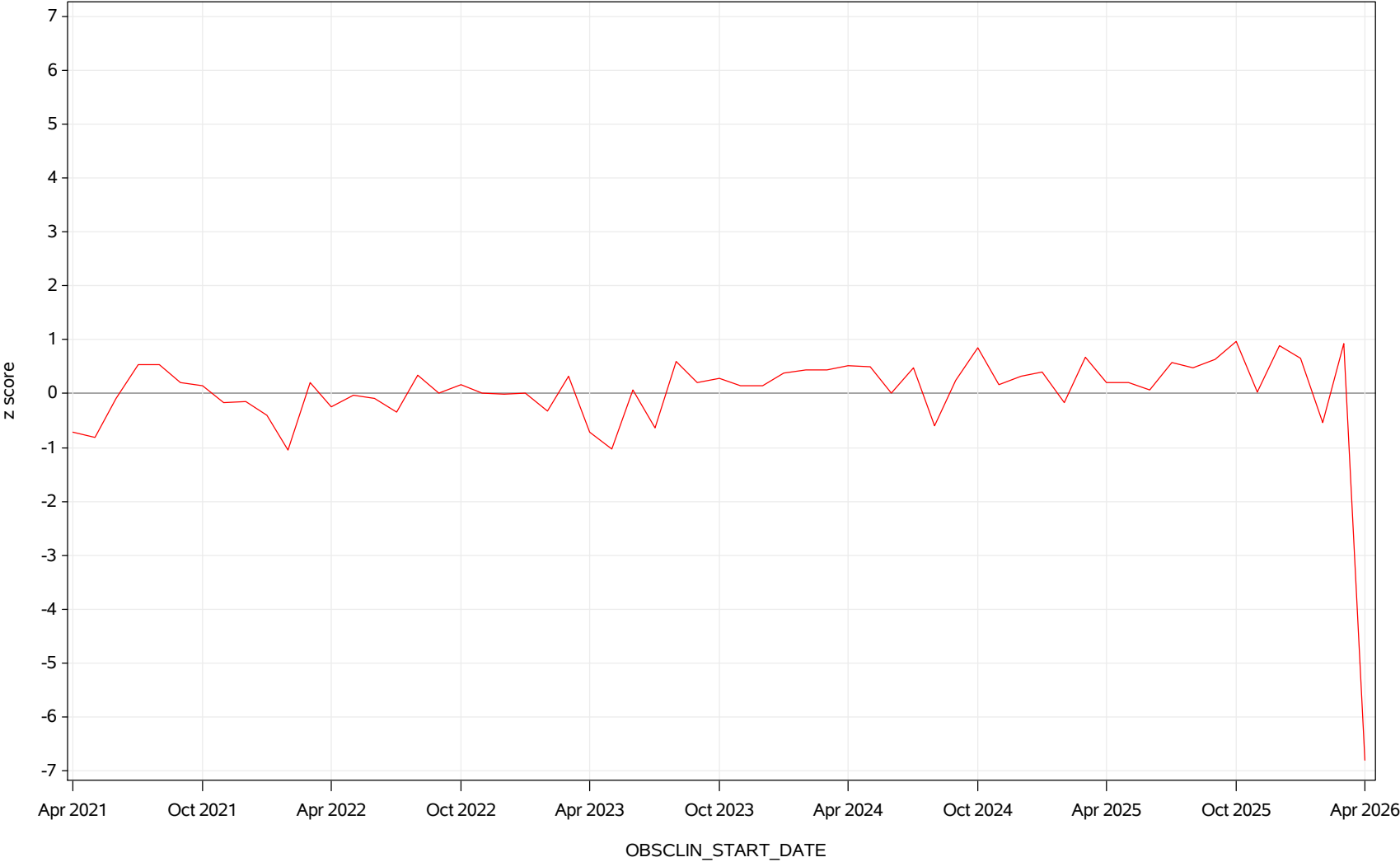


Chart IL. Trend in General Observation Records by Start Date, Past 5 Years

This chart illustrates relative changes over time in the number of records found in the OBS_GEN table. For all charts, the X-axis is the 60 months prior to the maximum refresh date. Monthly record counts were standardized over the period shown to have a mean of 0 and a standard deviation of 1. The y-axis reflects the deviation in each month's count from the mean. A value above 0 indicates an above-average number of records; a value below 0 indicates a below-average number of records. Significant inflection points and other unexpected patterns should be investigated.

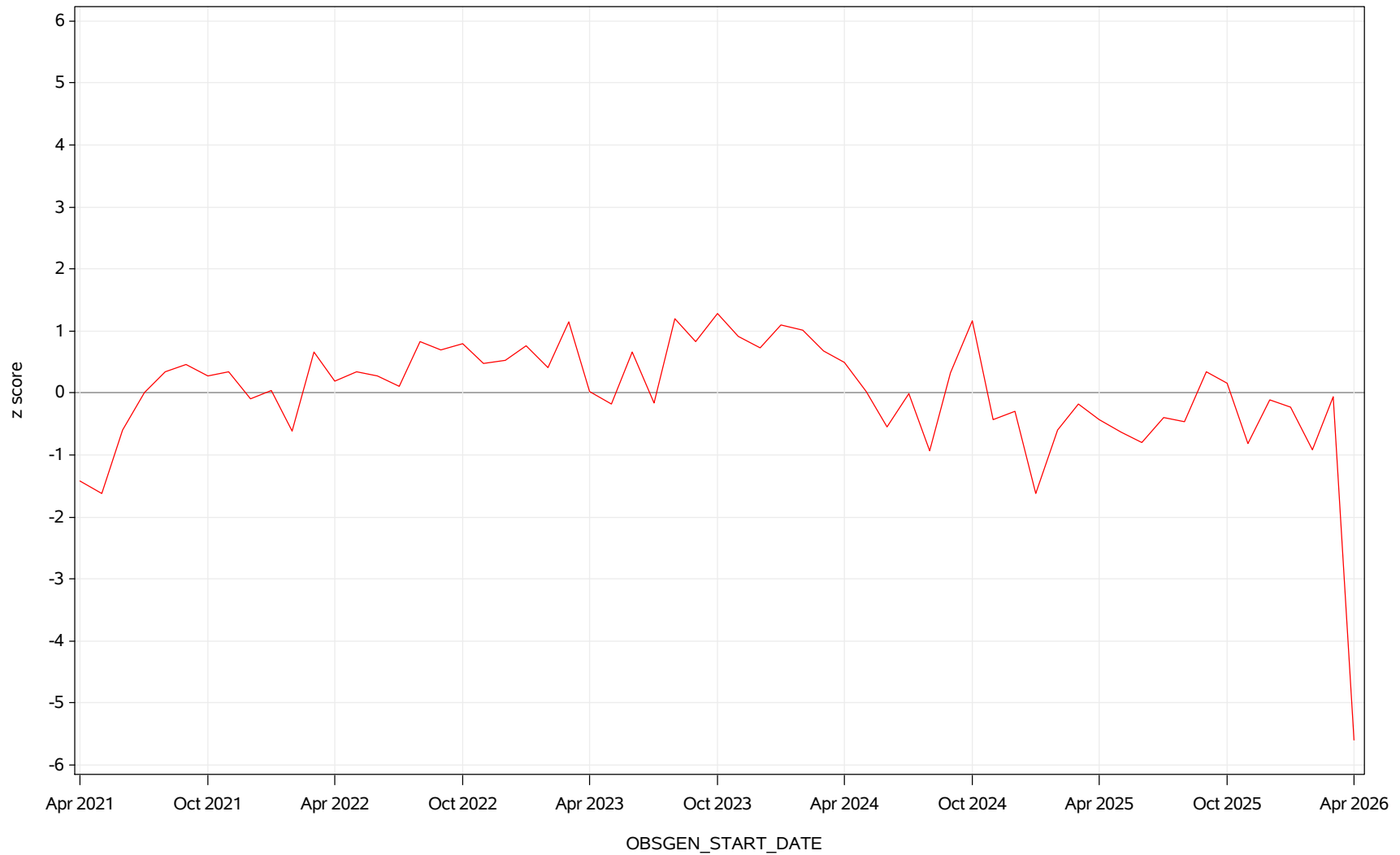


Table IIA. Primary Key Errors

This table shows the required primary key definitions and supports Data Check 1.05 (primary key definition errors). Data check exceptions are highlighted in red and must be corrected.

Table	CDM specifications for primary keys	Exception to specifications	Source table
CONDITION	CONDITIONID is unique	No	COND_L3_N
DEATH	DEATHID (concatenation of PATID and DEATH_SOURCE) is unique	No	DEATH_L3_N
DEATH_CAUSE	DEATHCID (concatenation of PATID + DEATH_CAUSE + DEATH_CAUSE_CODE + DEATH_CAUSE_TYPE + DEATH_CAUSE_SOURCE) is unique	No	DEATHC_L3_N
DEMOGRAPHIC	PATID is unique	No	DEM_L3_N
DIAGNOSIS	DIAGNOSISID is unique	No	DIA_L3_N
DISPENSING	DISPENSINGID is unique	No	DISP_L3_N
ENCOUNTER	ENCOUNTERID is unique	No	ENC_L3_N
ENROLLMENT	ENROLLID (concatenation of PATID + ENR_START_DATE+ENR_BASIS) is unique	No	ENR_L3_N
EXTERNAL_MEDS	EXTMEDID is unique	No	EXTMED_L3_N
HARVEST	NETWORKID+DATAMARTID is unique	No	XTBL_L3_METADATA
HASH_TOKEN	HASHID (concatenation of PATID + TOKEN_ENCRYPTION_KEY) is unique	No	HASH_L3_N
IMMUNIZATION	IMMUNIZATIONID is unique	No	IMMUNE_L3_N
LAB_HISTORY	LABHISTORYID is unique	No	LABHIST_L3_N
LAB_RESULT_CM	LAB_RESULT_CM_ID is unique	No	LAB_L3_N
LDS_ADDRESS_HISTORY	ADDRESSID is unique	No	LDSADRS_L3_N
MED_ADMIN	MEDADMINID is unique	No	MEDADM_L3_N
OBS_CLIN	OBSCLINID is unique	No	OBSCLIN_L3_N
OBS_GEN	OBSGENID is unique	No	OBSGEN_L3_N
PAT_RELATIONSHIP	PATRELID (concatenation of PATID_1 + PATID_2 + RELATIONSHIP_TYPE) is unique	No	PATREL_L3_N
PCORNET_TRIAL	TRIAL_KEY (concatenation of PATID + TRIALID + PARTICIPANTID) is unique	No	TRIAL_L3_N
PRESCRIBING	PRESCRIBINGID is unique	No	PRES_L3_N
PROCEDURES	PROCEDURESID is unique	No	PRO_L3_N
PROVIDER	PROVIDERID is unique	No	PROV_L3_N
PRO_CM	PRO_CM_ID is unique	No	PROCM_L3_N
VITAL	VITALID is unique	No	VIT_L3_N

The data checking logic is that the count of all records (ALL_N) and distinct records for the variable of interest (DISTINCT_N) must match.

Table IIB. Values Outside of CDM Specifications

This table lists all fields with pre-defined value sets and supports Data Check 1.06 (required fields contain values outside of data model specifications). Data check exceptions are highlighted in red and must be corrected.

Table	Field	Number of records with values outside of specifications	Source table
All fields conform to specifications			

Table IIC. Non-Permissible Missing Values

This table lists all fields which are required to be populated, as defined by the table constraints in the CDM, and supports Data Check 1.07 (required fields have non-permissible missing values). Data check exceptions are highlighted in red and must be corrected.

Table	Field	Number of records with missing values	Source table
All fields conform to specifications			

Table IID. Diagnostic Errors

This table illustrates exceptions to Data Checks 1.01 (required tables are not present), 1.02 (required tables are not populated), 1.03 (required fields are not present), 1.04 (required fields do not conform to data model specifications for data type, length, or name), 1.15 (fields with undefined lengths that are present in more than one table do not have harmonized field lengths[^]), Data Check 1.17 (Zip codes in the ENCOUNTER or LDS_ADDRESS_HISTORY table do not conform to expected values) and Data Check 1.21 (STATE_FIPS, COUNTY_FIPS and RUCA_ZIP fields do not conform to the expected content.) Data check exceptions are highlighted in red and must be corrected.

Data Check	Data Check Description	Exception	Table(s)	Field(s)	Source table(s)
1.01	Required tables are not present	Required table is not present. All tables must be present in an instantiation of the CDM	None	n/a	DATAMART_ALL
1.02	Expected tables are not populated	Table required to be populated is not populated. DEMOGRAPHIC, ENROLLMENT, ENCOUNTER, DIAGNOSIS, PROCEDURES, and HARVEST must be populated. If EHR data is included in the DataMart, LAB_RESULT_CM, PRESCRIBING and VITAL must also be populated.	None	n/a	DATAMART_ALL
1.03	Required fields are not present	Required numeric field is not present	None	None	DATAMART_ALL
1.03	Required fields are not present	Required character field is not present	None	None	DATAMART_ALL
1.04	Required fields do not conform to data model specifications for data type, length, or name	Required character field is numeric	None	None	DATAMART_ALL
1.04	Required fields do not conform to data model specifications for data type, length, or name	Required numeric field is character	None	None	DATAMART_ALL
1.04	Required fields do not conform to data model specifications for data type, length, or name	Required field is present but of unexpected length	None	None	DATAMART_ALL
1.15	Fields with undefined lengths that are present in more than one table do not have harmonized field lengths	Field lengths are not harmonized for one or more of the following fields: PATID, PATID_1, PATID_2, ENCOUNTERID, PRESCRIBINGID, PROCEDURESID, PROVIDERID, MEDADMIN_PROVIDERID, OBSGEN_PROVIDERID, OBSCLIN_PROVIDERID, RX_PROVIDERID, and VX_PROVIDERID.	None	None	DATAMART_ALL

Diagnostic checks are performed on all CDM tables and fields. Diagnostic checks ensure conformance to CDM v7.0 table structure, variable names, SAS variable lengths, and SAS data types. ^ To identify the differing field lengths, either open the datamart_all.sas7bdat dataset (in dmlocal) and review the field length attribute or run the CDM Diagnostic Query, remembering that the patient and provider identifier fields vary by table (PATID has aliases PATID_1 and PATID_2; PROVIDERID has aliases MEDADMIN_PROVIDERID, OBSGEN_PROVIDERID, OBSCLIN_PROVIDERID, RX_PROVIDERID, or VX_PROVIDERID).

Table IID. Diagnostic Errors (continued)

This table illustrates exceptions to Data Checks 1.01 (required tables are not present), 1.02 (required tables are not populated), 1.03 (required fields are not present), 1.04 (required fields do not conform to data model specifications for data type, length, or name), 1.15 (fields with undefined lengths that are present in more than one table do not have harmonized field lengths[^]), Data Check 1.17 (Zip codes in the ENCOUNTER or LDS_ADDRESS_HISTORY table do not conform to expected values) and Data Check 1.21 (STATE_FIPS, COUNTY_FIPS and RUCA_ZIP fields do not conform to the expected content.) Data check exceptions are highlighted in red and must be corrected.

Data Check	Data Check Description	Exception	Table(s)	Field(s)	Source table(s)
1.17	Zip codes in the ENCOUNTER or LDS_ADDRESS_HISTORY table do not conform to expected values	LDS_ADDRESS_HISTORY.ADDRESS_ZIP5, LDS_ADDRESS_HISTORY.ADDRESS_ZIP9 or ENCOUNTER.FACILITY_LOCATION contains alphabetical characters or does not have the expected number of digits.	None	None	ENC_L3_N; LDSADRS_L3_N
1.21	Geo codes in the LDS_ADDRESS_HISTORY table do not conform to expected values	LDS_ADDRESS_HISTORY.STATE_FIPS, LDS_ADDRESS_HISTORY.COUNTY_FIPS, or LDS_ADDRESS_HISTORY.RUCA_ZIP contains alphabetical characters or does not have the expected number of digits, or the first 2 digits of COUNTY_FIPS does not match STATE_FIPS.	None	None	LDSADRS_L3_N

Diagnostic checks are performed on all CDM tables and fields. Diagnostic checks ensure conformance to CDM v7.0 table structure, variable names, SAS variable lengths, and SAS data types. [^] To identify the differing field lengths, either open the datamart_all.sas7bdat dataset (in dmlocal) and review the field length attribute or run the CDM Diagnostic Query, remembering that the patient and provider identifier fields vary by table (PATID has aliases PATID_1 and PATID_2; PROVIDERID has aliases MEDADMIN_PROVIDERID, OBSGEN_PROVIDERID, OBSCLIN_PROVIDERID, RX_PROVIDERID, or VX_PROVIDERID).

Table IIE. Orphan Records, Replication Errors, Encounter Duplication and Hash Token Duplication

This table illustrates exceptions to Data Checks 1.08 (tables contain orphan PATIDs), 1.09 (tables contain orphan ENCOUNTERIDs for more than 5% of records), 1.10 (replication errors between the ENCOUNTER, PROCEDURES and DIAGNOSIS tables), 1.11 (more than 5% of encounters are assigned to more than one patient), 1.12 (tables contain orphan PROVIDERIDs), 1.14 (patients are missing from HASH_TOKEN), and 1.19 (more than 10% of hash tokens are assigned to multiple patients). Orphan PATIDs are not present in the DEMOGRAPHIC table. Orphan ENCOUNTERIDs are not present in the ENCOUNTER table. Orphan PROVIDERIDs are not present in the PROVIDER table. Replication errors are ENCOUNTERIDs in the DIAGNOSIS or PROCEDURES table where the encounter type or admit date does not match the corresponding value in the ENCOUNTER table. Data check exceptions to 1.14 and 1.19 are highlighted in blue and must be explained in the ETL ADD; all other data check exceptions are highlighted in red and must be corrected.

Data Check	Data Check Description	Exception	Table(s)	Field(s)	Count	%	Source table(s)
1.08	Orphan PATIDs	Orphan PATID(S) in the CONDITION, DIAGNOSIS, DEATH, DEATH_CAUSE, DISPENSING, ENCOUNTER, ENROLLMENT, HASH_TOKEN, IMMUNIZATION, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, EXTERNAL_MEDS, MED_ADMIN, OBS_CLIN, OBS_GEN, PCORNET_TRIAL, PAT_RELATIONSHIP, PRESCRIBING,PROCEDURES, PRO_CM, or VITAL table		None			XTBL_L3_MISMATCH; ENR_L3_N; ENC_L3_N; DIA_L3_N; PRO_L3_N; VIT_L3_N; LAB_L3_N; PRES_L3_N; DISP_L3_N; DEATH_L3_N; DEATHC_L3_N; COND_L3_N; PROCM_L3_N; TRIAL_L3_N; OBSCLIN_L3_N; OBSGEN_L3_N; MEDADM_L3_N; HASH_L3_N; IMMUNE_L3_N; LDSADRS_L3_N EXTMED_L3_N PATREL_L3_N

For Data Checks 1.08, 1.09, 1.11 and 1.12, the count is the number of distinct IDs with an exception. For Data Check 1.10, the count is the number of records with replication errors. For Data Check 1.19, the count is the number of valid records (i.e. where the token is not an error code) where the tokens are not distinct (calculated as VALID_N minus VALID_DISTINCT_N).

Percent is calculated as the number of distinct IDs with exceptions (shown) divided by the total distinct IDs in the table (not shown). For example, if the DIAGNOSIS table has 2 orphan PATIDs (source table: XTBL_L3_MISMATCH) and 100 distinct PATIDs (source table: DEM_L3_N), the percentage of distinct IDs is 2.0%. For Data Check 1.19, the percent is calculated as 100%-valid_distinct_n_pct.

Source tables include all potential source tables for the data check calculations.

Table IIE. Orphan Records, Replication Errors, Encounter Duplication and Hash Token Duplication (continued - page 2 of 4)

This table illustrates exceptions to Data Checks 1.08 (tables contain orphan PATIDs), 1.09 (tables contain orphan ENCOUNTERIDs for more than 5% of records), 1.10 (replication errors between the ENCOUNTER, PROCEDURES and DIAGNOSIS tables), 1.11 (more than 5% of encounters are assigned to more than one patient), 1.12 (tables contain orphan PROVIDERIDs), 1.14 (patients are missing from HASH_TOKEN), and 1.19 (more than 10% of hash tokens are assigned to multiple patients). Orphan PATIDs are not present in the DEMOGRAPHIC table. Orphan ENCOUNTERIDs are not present in the ENCOUNTER table. Orphan PROVIDERIDs are not present in the PROVIDER table. Replication errors are ENCOUNTERIDs in the DIAGNOSIS or PROCEDURES table where the encounter type or admit date does not match the corresponding value in the ENCOUNTER table. Data check exceptions to 1.14 and 1.19 are highlighted in blue and must be explained in the ETL ADD; all other data check exceptions are highlighted in red and must be corrected.

Data Check	Data Check Description	Exception	Table(s)	Field(s)	Count	%	Source table(s)
1.09	Orphan ENCOUNTERIDs	Orphan ENCOUNTERID(S) for more than 5% of the records in the	PROCEDURES	ENCOUNTERID	1	0.0	XTBL_L3_MISMATCH;
		CONDITION, DIAGNOSIS,	VITAL	ENCOUNTERID	2	0.0	DIA_L3_N; PRO_L3_N;
		IMMUNIZATION, LAB_RESULT_CM,	PRESCRIBING	ENCOUNTERID	300	0.0	VIT_L3_N; PRES_L3_N;
		MED_ADMIN, OBS_CLIN, OBS_GEN,	PRO_CM	ENCOUNTERID	2	0.0	LAB_L3_N;
		PRESCRIBING, PROCEDURES,	MED_ADMIN	ENCOUNTERID	5	0.0	PROCM_L3_N;
		PRO_CM, or VITAL table	OBS_CLIN	ENCOUNTERID	8	0.0	COND_L3_N;
			OBS_GEN	ENCOUNTERID	597	0.0	OBSCLIN_L3_N;
							OBSGEN_L3_N;
							MEDADM_L3_N;
							IMMUNE_L3_N
1.10	Replication errors	Replication error(s) in ENC_TYPE		None		n/a	XTBL_L3_MISMATCH
		or ADMIT_DATE in the DIAGNOSIS				n/a	
		or PROCEDURES table				n/a	

For Data Checks 1.08, 1.09, 1.11 and 1.12, the count is the number of distinct IDs with an exception. For Data Check 1.10, the count is the number of records with replication errors. For Data Check 1.19, the count is the number of valid records (i.e. where the token is not an error code) where the tokens are not distinct (calculated as VALID_N minus VALID_DISTINCT_N).

Percent is calculated as the number of distinct IDs with exceptions (shown) divided by the total distinct IDs in the table (not shown). For example, if the DIAGNOSIS table has 2 orphan PATIDs (source table: XTBL_L3_MISMATCH) and 100 distinct PATIDs (source table: DEM_L3_N), the percentage of distinct IDs is 2.0%. For Data Check 1.19, the percent is calculated as 100%-valid_distinct_n_pct.

Source tables include all potential source tables for the data check calculations.

Table IIE. Orphan Records, Replication Errors, Encounter Duplication and Hash Token Duplication (continued - page 3 of 4)

This table illustrates exceptions to Data Checks 1.08 (tables contain orphan PATIDs), 1.09 (tables contain orphan ENCOUNTERIDs for more than 5% of records), 1.10 (replication errors between the ENCOUNTER, PROCEDURES and DIAGNOSIS tables), 1.11 (more than 5% of encounters are assigned to more than one patient), 1.12 (tables contain orphan PROVIDERIDs), 1.14 (patients are missing from HASH_TOKEN), and 1.19 (more than 10% of hash tokens are assigned to multiple patients). Orphan PATIDs are not present in the DEMOGRAPHIC table. Orphan ENCOUNTERIDs are not present in the ENCOUNTER table. Orphan PROVIDERIDs are not present in the PROVIDER table. Replication errors are ENCOUNTERIDs in the DIAGNOSIS or PROCEDURES table where the encounter type or admit date does not match the corresponding value in the ENCOUNTER table. Data check exceptions to 1.14 and 1.19 are highlighted in blue and must be explained in the ETL ADD; all other data check exceptions are highlighted in red and must be corrected.

Data Check	Data Check Description	Exception	Table(s)	Field(s)	Count	%	Source table(s)
1.11	More than 5% of encounters are assigned to more than one patient	An ENCOUNTERID in the CONDITION, DIAGNOSIS, ENCOUNTER, IMMUNIZATION, LAB_RESULT_CM, MED_ADMIN, OBS_CLIN, OBS_GEN, PRESCRIBING, PROCEDURES, PRO_CM, or VITAL table is associated with more than 1 PATID in the same table	PRESCRIBING OBS_CLIN OBS_GEN	ENCOUNTERID ENCOUNTERID ENCOUNTERID	19 2 25	0.0 0.0 0.0	XTBL_L3_NON_UNIQUE; COND_L3_N; DIA_L3_N; ENC_L3_N; LAB_L3_N; PRES_L3_N; PRO_L3_N; VIT_L3_N; PROCM_L3_N; OBSCLIN_L3_N; OBSGEN_L3_N; MEDADM_L3_N; IMMUNE_L3_N
1.12	Orphan PROVIDERIDs	Orphan PROVIDERID(S) in the ENCOUNTER, DIAGNOSIS, IMMUNIZATION, MED_ADMIN, OBS_CLIN, OBS_GEN, PRESCRIBING, or PROCEDURES table.		None			XTBL_L3_MISMATCH; DIA_L3_N; ENC_L3_N; MEDADM_L3_N; OBSCLIN_L3_N; OBSGEN_L3_N; PRES_L3_N; PRO_L3_N; IMMUNE_L3_N
1.14	A PATID in the DEMOGRAPHIC table is not included in the HASH_TOKEN table	A PATID in the DEMOGRAPHIC table is not included in the HASH_TOKEN table		None			XTBL_L3_MISMATCH

For Data Checks 1.08, 1.09, 1.11 and 1.12, the count is the number of distinct IDs with an exception. For Data Check 1.10, the count is the number of records with replication errors. For Data Check 1.19, the count is the number of valid records (i.e. where the token is not an error code) where the tokens are not distinct (calculated as VALID_N minus VALID_DISTINCT_N).

Percent is calculated as the number of distinct IDs with exceptions (shown) divided by the total distinct IDs in the table (not shown). For example, if the DIAGNOSIS table has 2 orphan PATIDs (source table: XTBL_L3_MISMATCH) and 100 distinct PATIDs (source table: DEM_L3_N), the percentage of distinct IDs is 2.0%. For Data Check 1.19, the percent is calculated as 100%-valid_distinct_n_pct.

Source tables include all potential source tables for the data check calculations.

Table IIE. Orphan Records, Replication Errors, Encounter Duplication and Hash Token Duplication (continued - page 4 of 4)

This table illustrates exceptions to Data Checks 1.08 (tables contain orphan PATIDs), 1.09 (tables contain orphan ENCOUNTERIDs for more than 5% of records), 1.10 (replication errors between the ENCOUNTER, PROCEDURES and DIAGNOSIS tables), 1.11 (more than 5% of encounters are assigned to more than one patient), 1.12 (tables contain orphan PROVIDERIDs), 1.14 (patients are missing from HASH_TOKEN), and 1.19 (more than 10% of hash tokens are assigned to multiple patients). Orphan PATIDs are not present in the DEMOGRAPHIC table. Orphan ENCOUNTERIDs are not present in the ENCOUNTER table. Orphan PROVIDERIDs are not present in the PROVIDER table. Replication errors are ENCOUNTERIDs in the DIAGNOSIS or PROCEDURES table where the encounter type or admit date does not match the corresponding value in the ENCOUNTER table. Data check exceptions to 1.14 and 1.19 are highlighted in blue and must be explained in the ETL ADD; all other data check exceptions are highlighted in red and must be corrected.

Data Check	Data Check Description	Exception	Table(s)	Field(s)	Count	%	Source table(s)
1.19	More than 10% of hash tokens are assigned to multiple patients	More than 10% of the values in any hash token field are assigned to multiple patients		None			HASH_L3_N

For Data Checks 1.08, 1.09, 1.11 and 1.12, the count is the number of distinct IDs with an exception. For Data Check 1.10, the count is the number of records with replication errors. For Data Check 1.19, the count is the number of valid records (i.e. where the token is not an error code) where the tokens are not distinct (calculated as VALID_N minus VALID_DISTINCT_N).

Percent is calculated as the number of distinct IDs with exceptions (shown) divided by the total distinct IDs in the table (not shown). For example, if the DIAGNOSIS table has 2 orphan PATIDs (source table: XTBL_L3_MISMATCH) and 100 distinct PATIDs (source table: DEM_L3_N), the percentage of distinct IDs is 2.0%. For Data Check 1.19, the percent is calculated as 100%-valid_distinct_n_pct.

Source tables include all potential source tables for the data check calculations.

Table IIF. Potential Code Errors

This table illustrates exceptions to Data Check 1.13 (more than 5% of ICD, CPT, LOINC, RXCUI, or NDC codes do not conform to the expected length or content) and 1.16 (laboratory results or clinical observations are recorded in the wrong table). Results will be displayed for the code types observed in the data. Exceptions to Data Check 1.13 are highlighted in red and must be corrected. The CDM specification provides guidance on addressing potential code type errors (see General Implementation Guidance issue #5). Exceptions to Data Check 1.16 are highlighted in blue and must be explained in the ETL ADD.

Table	Code Type	Distinct Codes	Records	Records with Code Type Errors	Percent of Records with Code Type Errors (Data Check 1.13)	Records in the Wrong Table	Percentage of Records in the Wrong Table (Data Check 1.16)
CONDITION	10	1305	5,443	0	0.00	0	0.00
DIAGNOSIS	09	13590	27,964,996	2,675	0.01	0	0.00
DIAGNOSIS	10	45646	55,662,618	5	0.00	0	0.00
DISPENSING	ND	25544	5,280,863	7,068	0.13	0	0.00
EXTERNAL_MEDS	RX	32355	7,427,821	0	0.00	0	0.00
IMMUNIZATION	CX	105	4,213,552	0	0.00	0	0.00
LAB_HISTORY	LC	1327	17,694	0	0.00	0	0.00
LAB_RESULT_CM	LC	2216	212,426,833	13,969	0.01	0	0.00
MED_ADMIN	ND	1455	50,319,777	0	0.00	0	0.00
OBS_CLIN	LC	245	744,764,460	0	0.00	0	0.00
OBS_GEN	LC	716	85,105,018	0	0.00	0	0.00
PRESCRIBING	RX	52339	111,593,504	0	0.00	0	0.00
PROCEDURES	09	3280	1,798,833	0	0.00	0	0.00
PROCEDURES	10	14790	560,775	0	0.00	0	0.00
PROCEDURES	CH	13968	105,159,360	16,215	0.02	0	0.00
PRO_CM	LC	58	5,517,612	0	0.00	0	0.00

Results are produced by the potential_code_errors program. If errors are present, see the Potential Code Errors PDF and the SAS datasets produced by this program (see the Potential Code Errors section of the Data Curation Query Workplan for a list of the datasets).

Table IIG. LOINC Panel Codes

This table illustrates exceptions to Data Check 1.20 (more than 5% of LOINC records in the LAB_RESULT_CM, PRO_CM, and OBS_CLIN tables are panel codes). Exceptions are highlighted in blue and must be explained in the ETL ADD.

Table	Field	LOINC records with a Panel_Type of 'Panel'	LOINC records	% of LOINC records that are panels	Source table(s)
LAB_RESULT_CM	LAB_LOINC	0	167,597,234	0.00	LAB_L3_LOINC
OBS_CLIN	OBSCLIN_CODE	4,664,901	486,423,175	0.96	OBSCLIN_L3_LOINC
PRO_CM	PRO_CODE	0	4,926,370	0.00	PROCM_L3_LOINC

Table IIIA. Future Dates

This table includes most but not all CDM date fields and supports Data Check 2.01 (more than 5% of records have future dates). Future dates are calculated as those with dates occurring after the maximum refresh date. Future dates may be attributable to data entry errors in the source data or ETL errors such as incorrect REFRESH dates or including scheduled appointments in the ENCOUNTER table. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with future dates			Source table(s)
		Numerator	Denominator	%	
DEMOGRAPHIC	BIRTH_DATE	0	2,275,371	0.00	XTBL_L3_DATES
ENROLLMENT	ENR_START_DATE	12	830,128	0.00	XTBL_L3_DATES
DEATH	DEATH_DATE	0	454,934	0.00	XTBL_L3_DATES
ENCOUNTER	ADMIT_DATE	0	15,658,577	0.00	XTBL_L3_DATES
ENCOUNTER	DISCHARGE_DATE	0	15,158,825	0.00	XTBL_L3_DATES
DIAGNOSIS	ADMIT_DATE	0	53,705,073	0.00	XTBL_L3_DATES
DIAGNOSIS	DX_DATE	0	53,679,045	0.00	XTBL_L3_DATES
PROCEDURES	ADMIT_DATE	1	58,774,019	0.00	XTBL_L3_DATES
PROCEDURES	PX_DATE	0	58,750,615	0.00	XTBL_L3_DATES
VITAL	MEASURE_DATE	0	36,127,241	0.00	XTBL_L3_DATES
PRESCRIBING	RX_ORDER_DATE	0	79,571,202	0.00	XTBL_L3_DATES
PRESCRIBING	RX_START_DATE	28,876	79,571,202	0.04	XTBL_L3_DATES
DISPENSING	DISPENSE_DATE	1	4,824,060	0.00	XTBL_L3_DATES
LAB_RESULT_CM	LAB_ORDER_DATE	0	167,597,234	0.00	XTBL_L3_DATES
LAB_RESULT_CM	SPECIMEN_DATE	0	166,825,599	0.00	XTBL_L3_DATES
LAB_RESULT_CM	RESULT_DATE	0	167,597,234	0.00	XTBL_L3_DATES
HARVEST	REFRESH_MAX	0			XTBL_L3_METADATA
CONDITION	REPORT_DATE	0	2,791,908	0.00	XTBL_L3_DATES
CONDITION	RESOLVE_DATE	119	152,770	0.08	XTBL_L3_DATES
CONDITION	ONSET_DATE	10	133,220	0.01	XTBL_L3_DATES
PRO_CM	PRO_DATE	0	29,359,922	0.00	XTBL_L3_DATES
MED_ADMIN	MEDADMIN_START_DATE	0	38,479,890	0.00	XTBL_L3_DATES
MED_ADMIN	MEDADMIN_STOP_DATE	0	38,479,890	0.00	XTBL_L3_DATES

The numerator is the number of records with dates after the maximum DataMart refresh date (see REFRESH_MAX in the XTBL_L3_METADATA table). The denominator is the number of records with a populated date.

Table IIIA. Future Dates (continued)

This table includes most but not all CDM date fields and supports Data Check 2.01 (more than 5% of records have future dates). Future dates are calculated as those with dates occurring after the maximum refresh date. Future dates may be attributable to data entry errors in the source data or ETL errors such as incorrect REFRESH dates or including scheduled appointments in the ENCOUNTER table. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with future dates			Source table(s)
		Numerator	Denominator	%	
OBS_CLIN	OBSCLIN_START_DATE	0	486,423,175	0.00	XTBL_L3_DATES
OBS_CLIN	OBSCLIN_STOP_DATE	0	486,423,175	0.00	XTBL_L3_DATES
OBS_GEN	OBSGEN_START_DATE	0	87,218,390	0.00	XTBL_L3_DATES
OBS_GEN	OBSGEN_STOP_DATE	0	87,218,390	0.00	XTBL_L3_DATES
LDS_ADDRESS_HISTORY	ADDRESS_PERIOD_START	0	2,852,571	0.00	XTBL_L3_DATES
LDS_ADDRESS_HISTORY	ADDRESS_PERIOD_END	0	970,864	0.00	XTBL_L3_DATES
IMMUNIZATION	VX_RECORD_DATE	0	2,436,968	0.00	XTBL_L3_DATES
IMMUNIZATION	VX_ADMIN_DATE	0	2,436,968	0.00	XTBL_L3_DATES
EXTERNAL_MEDS	EXT_RECORD_DATE	0	4,599,717	0.00	XTBL_L3_DATES
EXTERNAL_MEDS	EXT_PAT_START_DATE	0	0		XTBL_L3_DATES
EXTERNAL_MEDS	EXT_PAT_END_DATE	0	0		XTBL_L3_DATES
EXTERNAL_MEDS	EXT_END_DATE	232	2,478,065	0.01	XTBL_L3_DATES
PAT_RELATIONSHIP	RELATIONSHIP_START	0	0		XTBL_L3_DATES
PAT_RELATIONSHIP	RELATIONSHIP_END	0	0		XTBL_L3_DATES

The numerator is the number of records with dates after the maximum DataMart refresh date (see REFRESH_MAX in the XTBL_L3_METADATA table). The denominator is the number of records with a populated date.

Table IIIB. Records With Extreme Values

This table supports Data Check 2.02 (more than 10% of records fall into the lowest or highest categories of age, height, weight, diastolic blood pressure, systolic blood pressure, or dispensed days supply). A high percentage of records in these categories may signal incorrect measurement units. Exceptions for blood pressure measures are expected for primarily pediatric populations. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Data Check Parameters		Records	Records with values in the lowest category		Records with values in the highest category		Median	Source table
		Low	High		N	%	N	%		
DEMOGRAPHIC	AGE (derived from BIRTH_DATE)	<0 yrs.	>89 yrs.	2,275,371	153	0.0	106,158	4.7	51	DEM_L3_AGEYRSDIST2; DEM_L3_AGEYRSDIST1
VITAL	HT	<21 inches	>76 inches	7,510,913	98,397	1.3	44,225	0.6	66	VIT_L3_HT; VIT_L3_HT_DIST
VITAL	WT	<0 lbs.	>350 lbs.	10,227,034	5	0.0	141,342	1.4	173	VIT_L3_WT; VIT_L3_WT_DIST
VITAL	DIASTOLIC	<40 mgHg	>120 mgHg	29,223,534	321,938	1.1	136,445	0.5	n/a	VIT_L3_DIASTOLIC
VITAL	SYSTOLIC	<40 mgHg	>210 mgHg	29,181,144	10,209	0.0	49,149	0.2	n/a	VIT_L3_SYSTOLIC
DISPENSING	DISPENSE_SUP_GROUP	<1 day	>90 days	4,824,060	8,866	0.2	19,433	0.4	n/a	DISP_L3_SUPDIST2

Table excludes records with values outside of CDM specifications.

Table IIIC. Illogical Dates

This table shows the number of patients who have dates of service in the DEATH, DISPENSING, ENCOUNTER, LAB_RESULT_CM, PRESCRIBING, PROCEDURES, VITAL, MED_ADMIN, OBS_CLIN, or OBS_GEN table which occur before their birth date or after their death date; the number of patients who have procedure dates occurring more than 5 days before the admit date or more than five days after the discharge date for the same encounter in the PROCEDURES table; and the number of patients who have a start date before the end date in the OBS_CLIN and OBS_GEN tables. The table shows these numbers as a percentage of patients in the ENCOUNTER table. Although some patients may not be included in the ENCOUNTER table, patients with encounters are the most relevant denominator for this table. This table supports Data Check 2.03 (more than 5% of patients have illogical date relationships). Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

DATE_COMPARISON	Patients	Percentage of total patients in the ENCOUNTER table	Source tables
ADMIT_DATE < BIRTH_DATE	5,456	0.7	XTBL_L3_DATE_LOGIC;ENC_L3_N
DISCHARGE_DATE < BIRTH_DATE	95	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
PX_DATE < BIRTH_DATE	5,795	0.7	XTBL_L3_DATE_LOGIC;ENC_L3_N
DX_DATE < BIRTH_DATE	3,426	0.4	XTBL_L3_DATE_LOGIC;ENC_L3_N
MEASURE_DATE < BIRTH_DATE	267	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
DISPENSE_DATE < BIRTH_DATE	6	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
RX_START_DATE < BIRTH_DATE	1,979	0.2	XTBL_L3_DATE_LOGIC;ENC_L3_N
RESULT_DATE < BIRTH_DATE	149	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
DEATH_DATE < BIRTH_DATE	187	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
MEDADMIN_START_DATE < BIRTH_DATE	36	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
OBSCLIN_START_DATE < BIRTH_DATE	349	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
OBSGEN_START_DATE < BIRTH_DATE	8,643	1.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
VX_RECORD_DATE < BIRTH_DATE	165	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
ADMIT_DATE > DEATH_DATE	2,306	0.3	XTBL_L3_DATE_LOGIC;ENC_L3_N
DISCHARGE_DATE > DEATH_DATE	2,377	0.3	XTBL_L3_DATE_LOGIC;ENC_L3_N
PX_DATE > DEATH_DATE	1,995	0.2	XTBL_L3_DATE_LOGIC;ENC_L3_N
DX_DATE > DEATH_DATE	2,046	0.2	XTBL_L3_DATE_LOGIC;ENC_L3_N
MEASURE_DATE > DEATH_DATE	2,206	0.3	XTBL_L3_DATE_LOGIC;ENC_L3_N
DISPENSE_DATE > DEATH_DATE	718	0.1	XTBL_L3_DATE_LOGIC;ENC_L3_N
RX_START_DATE > DEATH_DATE	2,258	0.3	XTBL_L3_DATE_LOGIC;ENC_L3_N
RESULT_DATE > DEATH_DATE	1,511	0.2	XTBL_L3_DATE_LOGIC;ENC_L3_N
MEDADMIN_START_DATE > DEATH_DATE	1,555	0.2	XTBL_L3_DATE_LOGIC;ENC_L3_N
OBSCLIN_START_DATE > DEATH_DATE	2,225	0.3	XTBL_L3_DATE_LOGIC;ENC_L3_N

Table IIIC. Illogical Dates (continued)

This table shows the number of patients who have dates of service in the DEATH, DISPENSING, ENCOUNTER, LAB_RESULT_CM, PRESCRIBING, PROCEDURES, VITAL, MED_ADMIN, OBS_CLIN, or OBS_GEN table which occur before their birth date or after their death date; the number of patients who have procedure dates occurring more than 5 days before the admit date or more than five days after the discharge date for the same encounter in the PROCEDURES table; and the number of patients who have a start date before the end date in the OBS_CLIN and OBS_GEN tables. The table shows these numbers as a percentage of patients in the ENCOUNTER table. Although some patients may not be included in the ENCOUNTER table, patients with encounters are the most relevant denominator for this table. This table supports Data Check 2.03 (more than 5% of patients have illogical date relationships). Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

DATE_COMPARISON	Patients	Percentage of total patients in the ENCOUNTER table	Source tables
OBSGEN_START_DATE > DEATH_DATE	3,672	0.4	XTBL_L3_DATE_LOGIC;ENC_L3_N
VX_RECORD_DATE > DEATH_DATE	949	0.1	XTBL_L3_DATE_LOGIC;ENC_L3_N
ADMIT_DATE > DISCHARGE_DATE	0	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
PX_DATE is More Than 5 Days Prior To The ADMIT_DATE	20,672	2.5	XTBL_L3_DATE_LOGIC;ENC_L3_N
PX_DATE is More Than 5 Days After The DISCHARGE_DATE	20,046	2.4	XTBL_L3_DATE_LOGIC;ENC_L3_N
DX_DATE is More Than 5 Days Prior To The ADMIT_DATE	478	0.1	XTBL_L3_DATE_LOGIC;ENC_L3_N
DX_DATE is More Than 5 Days After The DISCHARGE_DATE	0	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
OBSCLIN_START_DATE > OBSCLIN_STOP_DATE	335	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
OBSGEN_START_DATE > OBSGEN_STOP_DATE	7,274	0.9	XTBL_L3_DATE_LOGIC;ENC_L3_N
RELATIONSHIP_START > RELATIONSHIP_END	0	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
EXT_PAT_START_DATE > EXT_PAT_END_DATE	0	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N
MEDADMIN_START_DATE > MEDADMIN_STOP_DATE	5	0.0	XTBL_L3_DATE_LOGIC;ENC_L3_N

Table IIID. Encounters Per Visit and Per Patient

This table shows the number of encounters, patients, encounters per patient, visits (unique encounters per patient, provider, encounter type, and day), and encounters per visit by encounter type. These data support Data Check 2.04 (the average number of encounters per visit is > 2.0 for inpatient (IP), emergency department (ED), or ED to inpatient (EI) encounters). A high number of encounters per visit may signal potential redundancy or duplication. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Encounter Type	Encounters	Patients	Encounters per Patient	Encounters with known PROVIDERID	Visit (unique combinations of PATID, ENC_TYPE, ADMIT_DATE, and PROVIDERID)	Encounters with known PROVIDERID per visit	Source table
AV (Ambulatory Visit)	14,054,751	764,843	18.4	14,054,751	13,122,344	1.07	ENC_L3_ENCTYPE
ED (Emergency Dept)	830,269	282,532	2.9	830,269	814,885	1.02	ENC_L3_ENCTYPE
EI (ED to IP Stay)	176,236	100,411	1.8	176,236	176,220	1.00	ENC_L3_ENCTYPE
IC (Institutional Professional Consult)	0	0		0	0		ENC_L3_ENCTYPE
IP (Inpatient Hospital Stay)	151,292	113,395	1.3	151,292	151,097	1.00	ENC_L3_ENCTYPE
IS (Non-acute Institutional Stay)	0	0		0	0		ENC_L3_ENCTYPE
OA (Other Ambulatory Visit)	0	0		0	0		ENC_L3_ENCTYPE
OS (Observation Stay)	0	0		0	0		ENC_L3_ENCTYPE
TH (Telehealth)	446,029	120,609	3.7	446,029	443,975	1.00	ENC_L3_ENCTYPE
Missing, NI, UN or OT	0	0		0	0		ENC_L3_ENCTYPE

Table IIIG. Monthly Record Volume Outliers, Selected Domains

This table displays data check exceptions for Data Check 2.08 (the monthly volume of encounter, diagnosis, procedure, vital, prescribing, or laboratory records is an outlier). Encounter types of interest are AV (Ambulatory Visit), TH (telehealth), ED (Emergency Department), EI (Emergency Department Admit to Inpatient Hospital Stay), and IP (Inpatient Hospital Stay). The Evaluation Window[^] is shown in the table. Outliers are identified by computing a Difference Ratio^{^^} of each month compared to statistics of the previous 12 months. An outlier is defined as 0 records or a Difference Ratio of -7.0 or less. Months are excluded from the data check calculation if (a) the average record count during the previous 12 months was <500 or (b) the standard deviation during the previous 12 months was 0. If the exception occurs within the 60 months prior to the end of the Evaluation Window, the data check exception will be graphically compared to monthly record volumes in Charts IIIA through Chart IIIF. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Evaluation Window [^]	Encounter Type	Exception Month	Records	Average Records in the Previous 12 Months	Standard Deviation in the Previous 12 Months	Difference Ratio ^{^^}	Source table
	No Data Table contains a Data Check Exception							

[^] The Evaluation Window is determined as follows. The most recent twelve months are excluded because they are already evaluated by the data latency checks (Data Checks 3.07 and 3.11). Months prior to the 5th percentile for the Encounter Admission Date, Rx Order Date, Lab Result Date, Med Admin Start Date, Vital Measurement Date, External Meds Record Date and Patient Reported Outcomes Response Date are excluded since volumes may be unstable in these early months.

^{^^} The Difference Ratio is calculated as the volume in the current month minus the average volume in the previous 12 months divided by the standard deviation during the previous 12 months.

Table IVA. Diagnosis Records Per Encounter and Per Patient, Overall and by Encounter Type

This table supports Data Check 3.01 (the average number of diagnoses records with known diagnosis types per encounter is below threshold [1.0 for ambulatory (AV), inpatient (IP), emergency department (ED), ED to inpatient (EI) encounters, or telehealth (TH) encounters]). Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Encounter Type	DIAGNOSIS records	DIAGNOSIS records with known DX_TYPE	ENCOUNTER records	Diagnosis records per encounter	Diagnosis records with known DX_TYPE per encounter	Source table
AV (Ambulatory Visit)	39,314,798	39,310,662	14,054,751	2.80	2.80	DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
ED (Emergency Dept)	4,348,678	4,348,654	830,269	5.24	5.24	DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
EI (ED to IP Stay)	5,546,492	5,527,951	176,236	31.47	31.37	DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
IC (Institutional Professional Consult)	0	0	0			DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
IP (Inpatient Hospital Stay)	3,209,018	3,200,165	151,292	21.21	21.15	DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
IS (Non-acute Institutional Stay)	0	0	0			DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
OA (Other Ambulatory Visit)	0	0	0			DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
OS (Observation Stay)	0	0	0			DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
TH (Telehealth)	1,286,087	1,286,086	446,029	2.88	2.88	DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
Missing, NI, UN or OT	0	0	0			DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE
Total	53,705,073	53,673,518	15,658,577	3.43	3.43	DIA_L3_ENCTYPE; DIA_L3_DXTYPE_ENCTYPE; ENC_L3_ENCTYPE

The denominator comes from the ENCOUNTER table.

Chart IVA. Diagnosis Records Per Encounter by Admit Date and Encounter Type, Past 5 Years

This chart complements the information shown in Table IVA. This chart displays changes over time in the number of diagnosis codes per encounter in the ENCOUNTER table. The X-axis is the 60 months prior to the maximum refresh date. Significant inflection points and other unexpected patterns should be investigated.

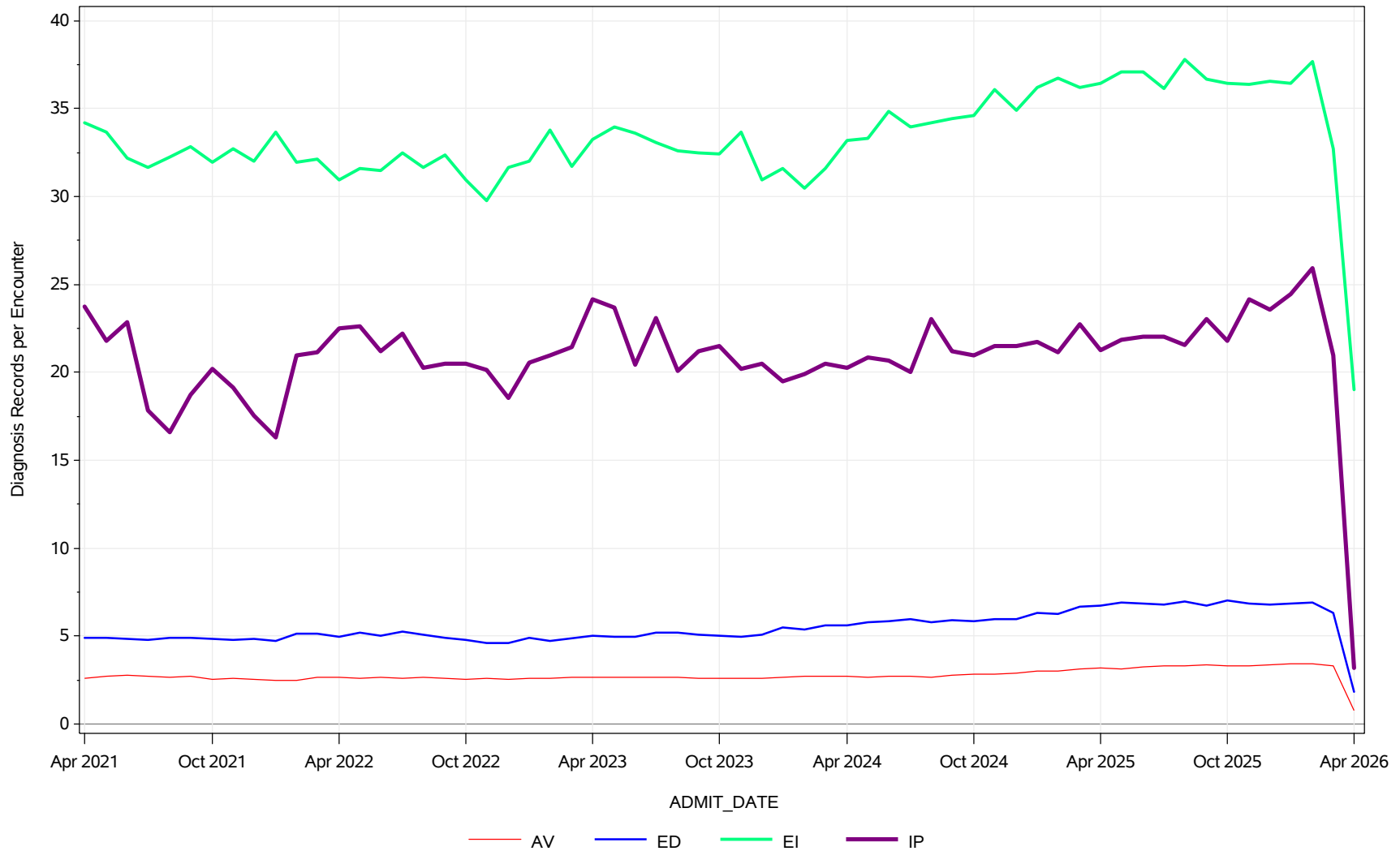


Table IVB. Procedure Records Per Encounter and Per Patient, Overall and by Encounter Type

This table supports Data Check 3.02 (the average number of procedure records with known procedure types per encounter is below threshold [0.75 for ambulatory (AV) encounters, 0.75 for emergency department (ED) encounters, 1.00 for ED to inpatient (EI) encounters, and 1.00 for inpatient (IP) encounters]). Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Encounter Type	PROCEDURES records	PROCEDURES records with known PX_TYPE	ENCOUNTER records	Procedures records per encounter	Procedures records with known PX_TYPE per encounter	Source table
AV (Ambulatory Visit)	39,945,453	39,915,295	14,054,751	2.84	2.84	PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
ED (Emergency Dept)	5,076,370	5,075,961	830,269	6.11	6.11	PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
EI (ED to IP Stay)	7,719,595	7,718,663	176,236	43.80	43.80	PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
IC (Institutional Professional Consult)	0	0	0			PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
IP (Inpatient Hospital Stay)	5,436,994	5,436,505	151,292	35.94	35.93	PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
IS (Non-acute Institutional Stay)	0	0	0			PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
OA (Other Ambulatory Visit)	0	0	0			PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
OS (Observation Stay)	0	0	0			PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
TH (Telehealth)	595,607	595,142	446,029	1.34	1.33	PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
Missing, NI, UN or OT	0	0	0			PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE
Total	58,774,019	58,741,566	15,658,577	3.75	3.75	PRO_L3_ENCTYPE; PRO_L3_PXTYPE_ENCTYPE; ENC_L3_ENCTYPE

The denominator comes from the ENCOUNTER table.

Chart IVB. Procedure Records Per Encounter by Admit Date and Encounter Type, Past 5 Years

This chart complements the information shown in Table IVB. This chart displays changes over time in the number of procedure codes per encounter in the ENCOUNTER table. The X-axis is the 60 months prior to the maximum refresh date. Significant inflection points and other unexpected patterns should be investigated.

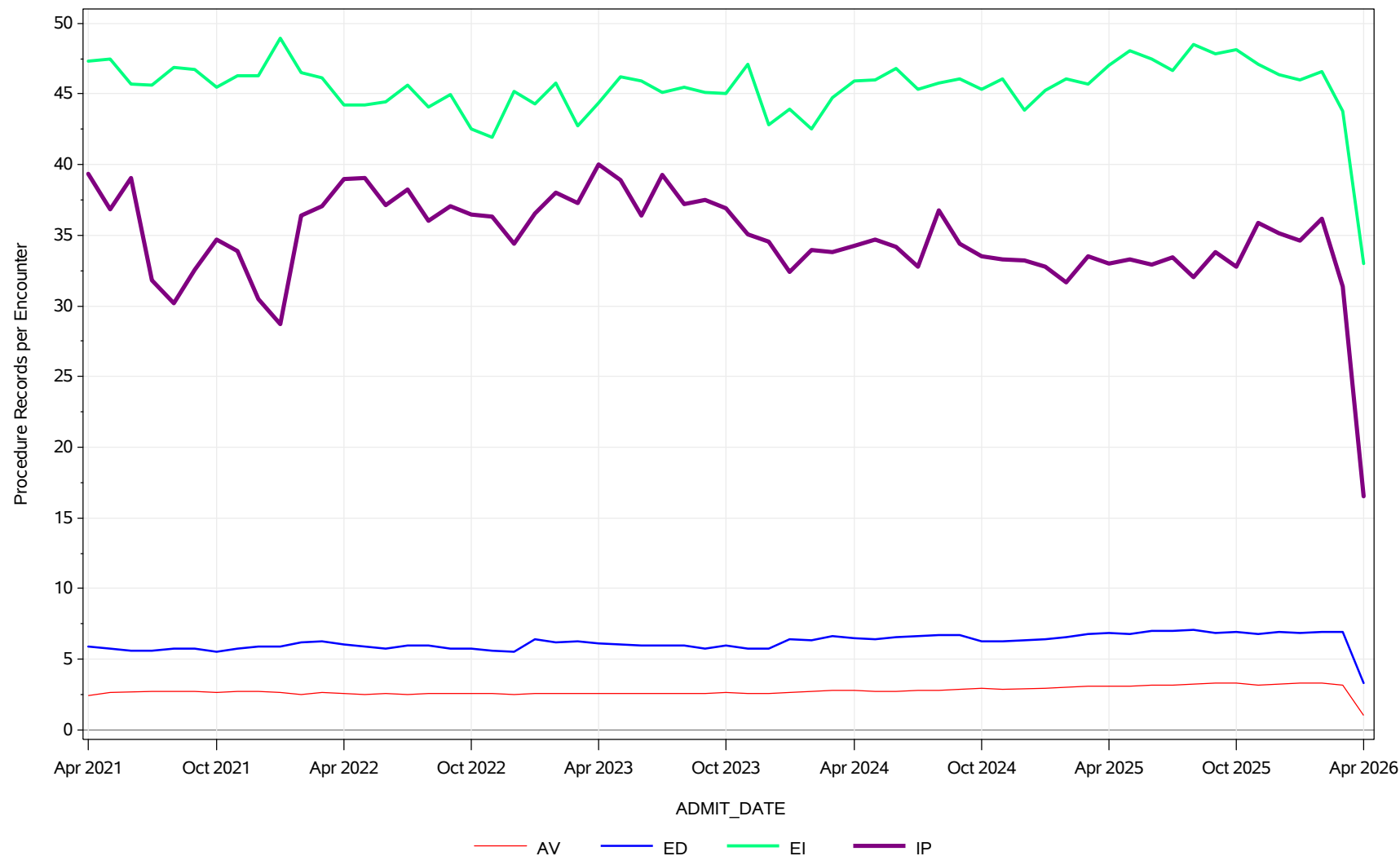


Table IVC. Missing or Unknown Values, Required Tables

This table includes fields in the DEMOGRAPHIC, DIAGNOSIS, ENCOUNTER, ENROLLMENT and PROCEDURES tables where the field is included in the query results and is not required to be populated. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: BIRTH_DATE, SEX, DISCHARGE_DATE (IP/EI encounters only), DISCHARGE_DISPOSITION (IP/EI encounters only), DX_ORIGIN, DX_SOURCE, DX_TYPE, DIAGNOSIS.ENCOUNTERID, PX_DATE, PX_SOURCE, PX_TYPE, and PROCEDURES.ENCOUNTERID. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Encounter Type Constraint	Records with missing, NI, UN, or OT values			Source table
			Numerator	Denominator	%	
DEMOGRAPHIC	BIOBANK_FLAG		2,284,557	2,284,557	100.0	DEM_L3_BIOBANK
DEMOGRAPHIC	BIRTH_DATE		9,186	2,284,557	0.4	XTBL_L3_DATES
DEMOGRAPHIC	BIRTH_TIME		9,186	2,284,557	0.4	XTBL_L3_TIMES
DEMOGRAPHIC	GENDER_IDENTITY		2,284,557	2,284,557	100.0	DEM_L3_GENDERDIST
DEMOGRAPHIC	HISPANIC		1,087,820	2,284,557	47.6	DEM_L3_HISPDIST
DEMOGRAPHIC	PAT_PREF_LANGUAGE_ SPOKEN		1,145,948	2,284,557	50.2	DEM_L3_PATPREFLANG
DEMOGRAPHIC	RACE		755,049	2,284,557	33.1	DEM_L3_RACEDIST
DEMOGRAPHIC	RACE_ETH_AI_AN		2,284,557	2,284,557	100.0	DEM_L3_RACEETHAIAN
DEMOGRAPHIC	RACE_ETH_ASIAN		2,284,557	2,284,557	100.0	DEM_L3_RACEETHASIAN
DEMOGRAPHIC	RACE_ETH_BLACK		2,284,557	2,284,557	100.0	DEM_L3_RACEETHBLACK
DEMOGRAPHIC	RACE_ETH_HISPANIC		2,284,557	2,284,557	100.0	DEM_L3_RACEETHHISPANIC
DEMOGRAPHIC	RACE_ETH_ME_NA		2,284,557	2,284,557	100.0	DEM_L3_RACEETHMENA
DEMOGRAPHIC	RACE_ETH_MISSING		2,284,557	2,284,557	100.0	DEM_L3_RACEETHMISS
DEMOGRAPHIC	RACE_ETH_NH_PI		2,284,557	2,284,557	100.0	DEM_L3_RACEETHNHPI
DEMOGRAPHIC	RACE_ETH_WHITE		2,284,557	2,284,557	100.0	DEM_L3_RACEETHWHITE
DEMOGRAPHIC	SEX		17,478	2,284,557	0.8	DEM_L3_SEXDIST
DEMOGRAPHIC	SEXUAL_ORIENTATION		2,284,557	2,284,557	100.0	DEM_L3_ORIENTDIST

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID) or the specific query for measures with encounter type constraints.

Table IVC. Missing or Unknown Values, Required Tables (continued - page 2 of 3)

This table includes fields in the DEMOGRAPHIC, DIAGNOSIS, ENCOUNTER, ENROLLMENT and PROCEDURES tables where the field is included in the query results and is not required to be populated. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: BIRTH_DATE, SEX, DISCHARGE_DATE (IP/EI encounters only), DISCHARGE_DISPOSITION (IP/EI encounters only), DX_ORIGIN, DX_SOURCE, DX_TYPE, DIAGNOSIS.ENCOUNTERID, PX_DATE, PX_SOURCE, PX_TYPE, and PROCEDURES.ENCOUNTERID. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Encounter Type Constraint	Records with missing, NI, UN, or OT values			Source table
			Numerator	Denominator	%	
DIAGNOSIS	ADMIT_DATE		0	53,705,073		XTBL_L3_DATES
DIAGNOSIS	DX_DATE		26,028	53,705,073	0.0	XTBL_L3_DATES
DIAGNOSIS	DX_ORIGIN		0	53,705,073		DIA_L3_ORIGIN
DIAGNOSIS	DX_POA		4,816	53,705,073	0.0	DIA_L3_DXPOA
DIAGNOSIS	DX_SOURCE		21,312	53,705,073	0.0	DIA_L3_DXSOURCE
DIAGNOSIS	DX_TYPE		31,555	53,705,073	0.1	DIA_L3_DXTYPE
DIAGNOSIS	ENCOUNTERID		0	53,705,073		DIA_L3_N
DIAGNOSIS	ENC_TYPE		0	53,705,073		DIA_L3_ENCTYPE
DIAGNOSIS	PDX	IP or EI	0	8,755,510		DIA_L3_PDX_ENCTYPE
DIAGNOSIS	PROVIDERID		2,385,053	53,705,073	4.4	DIA_L3_N
ENCOUNTER	ADMITTING_SOURCE	IP or EI	6,173	327,528	1.9	ENC_L3_ENCTYPE_ADMSRC
ENCOUNTER	ADMIT_TIME		0	15,658,577		XTBL_L3_TIMES
ENCOUNTER	DISCHARGE_DATE	IP or EI	1,015	327,528	0.3	ENC_L3_ENCTYPE_DDATE_YM
ENCOUNTER	DISCHARGE_DISPOSITION	IP or EI	1,326	327,528	0.4	ENC_L3_ENCTYPE_DISDISP
ENCOUNTER	DISCHARGE_STATUS	IP or EI	7,636	327,528	2.3	ENC_L3_ENCTYPE_DISSTAT
ENCOUNTER	DISCHARGE_TIME		499,751	15,658,577	3.2	XTBL_L3_TIMES
ENCOUNTER	DRG	IP or EI	18,715	327,528	5.7	ENC_L3_ENCTYPE_DRG
ENCOUNTER	DRG_TYPE		15,346,806	15,658,577	98.0	ENC_L3_DRG_TYPE
ENCOUNTER	ENC_TYPE		0	15,658,577		ENC_L3_ENCTYPE
ENCOUNTER	FACILITYID		914,699	15,658,577	5.8	ENC_L3_N
ENCOUNTER	FACILITY_LOCATION		0	15,658,577		ENC_L3_FACILITYLOC
ENCOUNTER	FACILITY_TYPE		15,658,577	15,658,577	100.0	ENC_L3_FACILITYTYPE

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID) or the specific query for measures with encounter type constraints.

Table IVC. Missing or Unknown Values, Required Tables (continued - page 3 of 3)

This table includes fields in the DEMOGRAPHIC, DIAGNOSIS, ENCOUNTER, ENROLLMENT and PROCEDURES tables where the field is included in the query results and is not required to be populated. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: BIRTH_DATE, SEX, DISCHARGE_DATE (IP/EI encounters only), DISCHARGE_DISPOSITION (IP/EI encounters only), DX_ORIGIN, DX_SOURCE, DX_TYPE, DIAGNOSIS.ENCOUNTERID, PX_DATE, PX_SOURCE, PX_TYPE, and PROCEDURES.ENCOUNTERID. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Encounter Type Constraint	Records with missing, NI, UN, or OT values			Source table
			Numerator	Denominator	%	
ENCOUNTER	PAYER_TYPE_PRIMARY		7,134,145	15,658,577	45.6	ENC_L3_PAYERTYPE1
ENCOUNTER	PAYER_TYPE_SECONDARY		977,279	15,658,577	6.2	ENC_L3_PAYERTYPE2
ENCOUNTER	PROVIDERID		0	15,658,577		ENC_L3_N
ENROLLMENT	CHART		0	830,128		ENR_L3_CHART
ENROLLMENT	ENR_END_DATE		0	830,128		XTBL_L3_DATES
PROCEDURES	ADMIT_DATE		0	58,774,019		XTBL_L3_DATES
PROCEDURES	ENCOUNTERID		0	58,774,019		PRO_L3_N
PROCEDURES	ENC_TYPE		0	58,774,019		PRO_L3_ENCTYPE
PROCEDURES	PPX		45,662,104	58,774,019	77.7	PRO_L3_PPX
PROCEDURES	PROVIDERID		78,194	58,774,019	0.1	PRO_L3_N
PROCEDURES	PX_DATE		23,404	58,774,019	0.0	XTBL_L3_DATES
PROCEDURES	PX_SOURCE		0	58,774,019		PRO_L3_PXSOURCE
PROCEDURES	PX_TYPE		32,453	58,774,019	0.1	PRO_L3_PXTYPE

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID) or the specific query for measures with encounter type constraints.

Table IVD. Missing or Unknown Values, Optional Tables

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
CONDITION	CONDITION_SOURCE	0	2,791,908		COND_L3_SOURCE
CONDITION	CONDITION_STATUS	0	2,791,908		COND_L3_STATUS
CONDITION	CONDITION_TYPE	750	2,791,908	0.0	COND_L3_TYPE
CONDITION	ENCOUNTERID	2,791,908	2,791,908	100.0	COND_L3_N
CONDITION	ONSET_DATE	2,658,688	2,791,908	95.2	XTBL_L3_DATES
CONDITION	REPORT_DATE	0	2,791,908		XTBL_L3_DATES
CONDITION	RESOLVE_DATE	2,639,138	2,791,908	94.5	XTBL_L3_DATES
DEATH	DEATH_DATE	0	454,934		XTBL_L3_DATES
DEATH	DEATH_DATE_IMPUTE	0	454,934		DEATH_L3_IMPUTE
DEATH	DEATH_MATCH_CONFIDENCE	0	454,934		DEATH_L3_MATCH
DEATH	DEATH_SOURCE	0	454,934		DEATH_L3_SOURCE
DEATH_CAUSE	DEATH_CAUSE_CODE	0	19,017		DEATHC_L3_CODE
DEATH_CAUSE	DEATH_CAUSE_CONFIDENCE	0	19,017		DEATHC_L3_CONF
DEATH_CAUSE	DEATH_CAUSE_SOURCE	0	19,017		DEATHC_L3_SOURCE
DEATH_CAUSE	DEATH_CAUSE_TYPE	0	19,017		DEATHC_L3_TYPE
DISPENSING	DISPENSE_AMT	0	4,824,060		DISP_L3_DISPAMT_DIST
DISPENSING	DISPENSE_DOSE_DISP	2,393,574	4,824,060	49.6	DISP_L3_DOSE_DIST
DISPENSING	DISPENSE_DOSE_DISP_UNIT	2,082,954	4,824,060	43.2	DISP_L3_DOSEUNIT

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVD. Missing or Unknown Values, Optional Tables (continued - page 2 of 9)

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
DISPENSING	DISPENSE_ROUTE	2,515,730	4,824,060	52.1	DISP_L3_ROUTE
DISPENSING	DISPENSE_SOURCE	0	4,824,060		DISP_L3_SOURCE
DISPENSING	DISPENSE_SUP	0	4,824,060		DISP_L3_SUPDIST2
DISPENSING	PRESCRIBINGID	4,824,060	4,824,060	100.0	DISP_L3_N
EXTERNAL_MEDS	EXTMED_SOURCE	0	4,599,717		EXTMED_L3_SOURCE
EXTERNAL_MEDS	EXT_BASIS	0	4,599,717		EXTMED_L3_BASIS
EXTERNAL_MEDS	EXT_DOSE_FORM	2,211,797	4,599,717	48.1	EXTMED_L3_RXDOSEFORM
EXTERNAL_MEDS	EXT_DOSE_ORDERED_UNIT	1,970,697	4,599,717	42.8	EXTMED_L3_RXDOSE_UNIT
EXTERNAL_MEDS	EXT_RECORD_DATE	0	4,599,717		EXTMED_L3_RDATE_Y
EXTERNAL_MEDS	EXT_ROUTE	1,038,077	4,599,717	22.6	EXTMED_L3_ROUTE
EXTERNAL_MEDS	RXNORM_CUI	412,830	4,599,717	9.0	EXTMED_L3_RXCUI
IMMUNIZATION	ENCOUNTERID	658,405	2,436,968	27.0	IMMUNE_L3_N
IMMUNIZATION	PROCEDURESID	2,285,450	2,436,968	93.8	IMMUNE_L3_N
IMMUNIZATION	VX_ADMIN_DATE	0	2,436,968		XTBL_L3_DATES
IMMUNIZATION	VX_BODY_SITE	1,345,572	2,436,968	55.2	IMMUNE_L3_BODYSITE
IMMUNIZATION	VX_CODE_TYPE	132,584	2,436,968	5.4	IMMUNE_L3_CODETYPE
IMMUNIZATION	VX_DOSE	0	2,436,968		IMMUNE_L3_DOSE_DIST
IMMUNIZATION	VX_DOSE_UNIT	776,354	2,436,968	31.9	IMMUNE_L3_DOSEUNIT

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVD. Missing or Unknown Values, Optional Tables (continued - page 3 of 9)

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
IMMUNIZATION	VX_EXP_DATE	1,323,473	2,436,968	54.3	XTBL_L3_DATES
IMMUNIZATION	VX_LOT_NUM	546,628	2,436,968	22.4	IMMUNE_L3_LOTNUM
IMMUNIZATION	VX_MANUFACTURER	1,305,224	2,436,968	53.6	IMMUNE_L3_MANUFACTURER
IMMUNIZATION	VX_PROVIDERID	2,234,635	2,436,968	91.7	IMMUNE_L3_N
IMMUNIZATION	VX_RECORD_DATE	0	2,436,968		XTBL_L3_DATES
IMMUNIZATION	VX_ROUTE	2,349,440	2,436,968	96.4	IMMUNE_L3_ROUTE
IMMUNIZATION	VX_SOURCE	0	2,436,968		IMMUNE_L3_SOURCE
IMMUNIZATION	VX_STATUS	33,154	2,436,968	1.4	IMMUNE_L3_STATUS
IMMUNIZATION	VX_STATUS_REASON	0	2,436,968		IMMUNE_L3_STATUSREASON
LAB_HISTORY	AGE_MAX_WKS	0	17,694		LABHIST_L3_MAX_WKS
LAB_HISTORY	AGE_MIN_WKS	0	17,694		LABHIST_L3_MIN_WKS
LAB_HISTORY	LAB_FACILITYID	0	17,694		LABHIST_L3_N
LAB_HISTORY	NORM_MODIFIER_HIGH	17,694	17,694	100.0	LABHIST_L3_HIGH
LAB_HISTORY	NORM_MODIFIER_LOW	17,694	17,694	100.0	LABHIST_L3_LOW
LAB_HISTORY	PERIOD_END	0	17,694		LABHIST_L3_PDEND_Y
LAB_HISTORY	PERIOD_START	0	17,694		LABHIST_L3_PDSTART_Y
LAB_HISTORY	RACE	17,694	17,694	100.0	LABHIST_L3_RACEDIST
LAB_HISTORY	RESULT_UNIT	14,087	17,694	79.6	LABHIST_L3_UNIT

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVD. Missing or Unknown Values, Optional Tables (continued - page 4 of 9)

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
LAB_HISTORY	SEX	12,264	17,694	69.3	LABHIST_L3_SEXDIST
LAB_RESULT_CM	ABN_IND	36,755,578	167,597,234	21.9	LAB_L3_ABN
LAB_RESULT_CM	ENCOUNTERID	0	167,597,234		LAB_L3_N
LAB_RESULT_CM	LAB_LOINC	17,296,092	167,597,234	10.3	LAB_L3_LOINC
LAB_RESULT_CM	LAB_LOINC_SOURCE	17,296,092	167,597,234	10.3	LAB_L3_LSOURCE
LAB_RESULT_CM	LAB_ORDER_DATE	0	167,597,234		XTBL_L3_DATES
LAB_RESULT_CM	LAB_PX	57,911,544	167,597,234	34.6	LAB_L3_PX_PXTYPE
LAB_RESULT_CM	LAB_PX_TYPE	0	167,597,234		LAB_L3_PX_TYPE
LAB_RESULT_CM	LAB_RESULT_SOURCE	0	167,597,234		LAB_L3_RSOURCE
LAB_RESULT_CM	NORM_MODIFIER_HIGH	0	167,597,234		LAB_L3_HIGH
LAB_RESULT_CM	NORM_MODIFIER_LOW	1,347,194	167,597,234	0.8	LAB_L3_LOW
LAB_RESULT_CM	PRIORITY	39,773,654	167,597,234	23.7	LAB_L3_PRIORITY
LAB_RESULT_CM	RESULT_LOC	0	167,597,234		LAB_L3_LOC
LAB_RESULT_CM	RESULT_MODIFIER	309,396	167,597,234	0.2	LAB_L3_MOD
LAB_RESULT_CM	RESULT_NUM	20,458,784	167,597,234	12.2	LAB_L3_LOINC_RESULT_NUM
LAB_RESULT_CM	RESULT_QUAL	153,344,884	167,597,234	91.5	LAB_L3_QUAL
LAB_RESULT_CM	RESULT_SNOMED	167,597,234	167,597,234	100.0	LAB_L3_SNOMED
LAB_RESULT_CM	RESULT_TIME	0	167,597,234		XTBL_L3_TIMES

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVD. Missing or Unknown Values, Optional Tables (continued - page 5 of 9)

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
LAB_RESULT_CM	RESULT_UNIT	45,421,718	167,597,234	27.1	LAB_L3_UNIT
LAB_RESULT_CM	SPECIMEN_DATE	771,635	167,597,234	0.5	XTBL_L3_DATES
LAB_RESULT_CM	SPECIMEN_SOURCE	991,322	167,597,234	0.6	LAB_L3_SOURCE
LAB_RESULT_CM	SPECIMEN_TIME	771,635	167,597,234	0.5	XTBL_L3_TIMES
LDS_ADDRESS_HISTORY	ADDRESS_CITY	3,834	2,852,571	0.1	LDSADRS_L3_ADRSCITY
LDS_ADDRESS_HISTORY	ADDRESS_COUNTY	765,478	2,852,571	26.8	LDSADRS_L3_N
LDS_ADDRESS_HISTORY	ADDRESS_PERIOD_END	1,881,707	2,852,571	66.0	XTBL_L3_DATES
LDS_ADDRESS_HISTORY	ADDRESS_PERIOD_START	0	2,852,571		XTBL_L3_DATES
LDS_ADDRESS_HISTORY	ADDRESS_STATE	0	2,852,571		LDSADRS_L3_ADRSSTATE
LDS_ADDRESS_HISTORY	ADDRESS_TYPE	2,610	2,852,571	0.1	LDSADRS_L3_ADRSTYPE
LDS_ADDRESS_HISTORY	ADDRESS_USE	0	2,852,571		LDSADRS_L3_ADRSUSE
LDS_ADDRESS_HISTORY	ADDRESS_ZIP5	18,598	2,852,571	0.7	LDSADRS_L3_ADRSZIP5
LDS_ADDRESS_HISTORY	ADDRESS_ZIP9	817,687	2,852,571	28.7	LDSADRS_L3_ADRSZIP9
LDS_ADDRESS_HISTORY	COUNTY_FIPS	765,478	2,852,571	26.8	LDSADRS_L3_COUNTYFIPS
LDS_ADDRESS_HISTORY	CURRENT_ADDRESS_FLAG	0	2,852,571		LDSADRS_L3_CURRENT
LDS_ADDRESS_HISTORY	RUCA_ZIP	25,417	2,852,571	0.9	LDSADRS_L3_RUCAZIP
LDS_ADDRESS_HISTORY	STATE_FIPS	6,822	2,852,571	0.2	LDSADRS_L3_STATEFIPS
MED_ADMIN	ENCOUNTERID	0	38,479,890		MEDADM_L3_N

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVD. Missing or Unknown Values, Optional Tables (continued - page 6 of 9)

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
MED_ADMIN	MEDADMIN_CODE	2,243,590	38,479,890	5.8	MEDADM_L3_CODE_TYPE
MED_ADMIN	MEDADMIN_DOSE_ADMIN	9,614,236	38,479,890	25.0	MEDADM_L3_DOSEADM
MED_ADMIN	MEDADMIN_DOSE_ADMIN_UNIT	9,609,250	38,479,890	25.0	MEDADM_L3_DOSEADMUNIT
MED_ADMIN	MEDADMIN_PROVIDERID	25,112,262	38,479,890	65.3	MEDADM_L3_N
MED_ADMIN	MEDADMIN_ROUTE	589,158	38,479,890	1.5	MEDADM_L3_ROUTE
MED_ADMIN	MEDADMIN_SOURCE	0	38,479,890		MEDADM_L3_SOURCE
MED_ADMIN	MEDADMIN_START_TIME	0	38,479,890		XTBL_L3_TIMES
MED_ADMIN	MEDADMIN_STOP_DATE	0	38,479,890		XTBL_L3_DATES
MED_ADMIN	MEDADMIN_STOP_TIME	0	38,479,890		XTBL_L3_TIMES
MED_ADMIN	MEDADMIN_TYPE	0	38,479,890		MEDADM_L3_TYPE
MED_ADMIN	PRESCRIBINGID	928	38,479,890	0.0	MEDADM_L3_N
OBS_CLIN	ENCOUNTERID	27,122,126	486,423,175	5.6	OBSCLIN_L3_N
OBS_CLIN	OBSCLIN_ABN_IND	482,346,063	486,423,175	99.2	OBSCLIN_L3_ABN
OBS_CLIN	OBSCLIN_CODE	0	486,423,175		OBSCLIN_L3_CODE_TYPE
OBS_CLIN	OBSCLIN_PROVIDERID	481,325,709	486,423,175	99.0	OBSCLIN_L3_N
OBS_CLIN	OBSCLIN_RESULT_MODIFIER	0	486,423,175		OBSCLIN_L3_MOD
OBS_CLIN	OBSCLIN_RESULT_QUAL	471,147,452	486,423,175	96.9	OBSCLIN_L3_QUAL
OBS_CLIN	OBSCLIN_RESULT_UNIT	246,000,721	486,423,175	50.6	OBSCLIN_L3_RUNIT

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVD. Missing or Unknown Values, Optional Tables (continued - page 7 of 9)

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
OBS_CLIN	OBSCLIN_SOURCE	0	486,423,175		OBSCLIN_L3_SOURCE
OBS_CLIN	OBSCLIN_START_TIME	0	486,423,175		XTBL_L3_TIMES
OBS_CLIN	OBSCLIN_STOP_DATE	0	486,423,175		XTBL_L3_DATES
OBS_CLIN	OBSCLIN_STOP_TIME	0	486,423,175		XTBL_L3_TIMES
OBS_CLIN	OBSCLIN_TYPE	0	486,423,175		OBSCLIN_L3_TYPE
OBS_GEN	ENCOUNTERID	32,066,130	87,218,390	36.8	OBSGEN_L3_N
OBS_GEN	OBSGEN_ABN_IND	87,218,390	87,218,390	100.0	OBSGEN_L3_ABN
OBS_GEN	OBSGEN_CODE	4,369,117	87,218,390	5.0	OBSGEN_L3_CODE_TYPE
OBS_GEN	OBSGEN_PROVIDERID	87,217,640	87,218,390	100.0	OBSGEN_L3_N
OBS_GEN	OBSGEN_RESULT_MODIFIER	834,021	87,218,390	1.0	OBSGEN_L3_MOD
OBS_GEN	OBSGEN_RESULT_QUAL	87,218,390	87,218,390	100.0	OBSGEN_L3_QUAL
OBS_GEN	OBSGEN_RESULT_UNIT	87,218,390	87,218,390	100.0	OBSGEN_L3_RUNIT
OBS_GEN	OBSGEN_SOURCE	0	87,218,390		OBSGEN_L3_SOURCE
OBS_GEN	OBSGEN_START_TIME	0	87,218,390		XTBL_L3_TIMES
OBS_GEN	OBSGEN_STOP_DATE	0	87,218,390		XTBL_L3_DATES
OBS_GEN	OBSGEN_STOP_TIME	0	87,218,390		XTBL_L3_TIMES
OBS_GEN	OBSGEN_TYPE	0	87,218,390		OBSGEN_L3_TYPE
PAT_RELATIONSHIP	RELATIONSHIP_END	82,582	82,582	100.0	XTBL_L3_DATES

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVD. Missing or Unknown Values, Optional Tables (continued - page 8 of 9)

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
PAT_RELATIONSHIP	RELATIONSHIP_START	82,582	82,582	100.0	XTBL_L3_DATES
PRESCRIBING	ENCOUNTERID	9,007,915	79,571,202	11.3	PRES_L3_N
PRESCRIBING	RXNORM_CUI	969,892	79,571,202	1.2	PRES_L3_RXCUI
PRESCRIBING	RX_BASIS	0	79,571,202		PRES_L3_BASIS
PRESCRIBING	RX_DAYS_SUPPLY	77,493,369	79,571,202	97.4	PRES_L3_SUPDIST2
PRESCRIBING	RX_DISPENSE_AS_WRITTEN	64,690,921	79,571,202	81.3	PRES_L3_DISPASWRTN
PRESCRIBING	RX_DOSE_FORM	50,488,754	79,571,202	63.5	PRES_L3_RXDOSEFORM
PRESCRIBING	RX_DOSE_ORDERED	31,431,013	79,571,202	39.5	PRES_L3_RXDOSEODR_DIST
PRESCRIBING	RX_DOSE_ORDERED_UNIT	31,530,968	79,571,202	39.6	PRES_L3_RXDOSEODRUNIT
PRESCRIBING	RX_END_DATE	1,727,791	79,571,202	2.2	XTBL_L3_DATES
PRESCRIBING	RX_FREQUENCY	38,899,397	79,571,202	48.9	PRES_L3_FREQ
PRESCRIBING	RX_ORDER_DATE	0	79,571,202		XTBL_L3_DATES
PRESCRIBING	RX_ORDER_TIME	0	79,571,202		XTBL_L3_TIMES
PRESCRIBING	RX_PRN_FLAG	0	79,571,202		PRES_L3_PRNFLAG
PRESCRIBING	RX_PROVIDERID	37,409,935	79,571,202	47.0	PRES_L3_N
PRESCRIBING	RX_QUANTITY	64,098,854	79,571,202	80.6	PRES_L3_RXQTY_DIST
PRESCRIBING	RX_REFILLS	1	79,571,202	0.0	PRES_L3_RXREFILL_DIST
PRESCRIBING	RX_ROUTE	15,122,993	79,571,202	19.0	PRES_L3_ROUTE

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVD. Missing or Unknown Values, Optional Tables (continued - page 9 of 9)

This table includes most of the fields in the CONDITION, DEATH, DEATH_CAUSE, DISPENSING, EXTERNAL_MEDS, IMMUNIZATION, LAB_HISTORY, LAB_RESULT_CM, LDS_ADDRESS_HISTORY, MED_ADMIN, OBS_CLIN, OBS_GEN, PAT_RELATIONSHIP, PRESCRIBING, PRO_CM, PROVIDER and VITAL tables where the field is included in the query results and is not required to be populated. Vital measures (e.g. HEIGHT) are not included because the table structure does not support missingness assessment. The table depicts the percentage of records with missing or unknown values. Results support Data Check 3.03 (more than 10% of records have missing or unknown values) for the following fields: RX_ORDER_DATE, VX_RECORD_DATE, DISPENSE_SUP, DEATH_SOURCE, DEATH_CAUSE_CODE, DEATH_CAUSE_SOURCE, DISPENSE_SOURCE, CONDITION_SOURCE, LAB_RESULT_SOURCE, MEDADMIN_SOURCE, RX_SOURCE, VITAL_SOURCE, VX_SOURCE, VITAL.ENCOUNTERID, MED_ADMIN.ENCOUNTERID, LAB_RESULT_CM.ENCOUNTERID, MEDADMIN_CODE, MEDADMIN_TYPE, OBSCLIN_CODE, OBSCLIN_TYPE, OBSGEN_CODE, OBSGEN_TYPE, EXT_RECORD_DATE, EXTMED_SOURCE and EXT_BASIS. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Field	Records with missing, NI, UN, or OT values			Source table
		Numerator	Denominator	%	
PRESCRIBING	RX_SOURCE	0	79,571,202		PRES_L3_SOURCE
PRESCRIBING	RX_START_DATE	0	79,571,202		XTBL_L3_DATES
PROVIDER	PROVIDER_NPI	6,618	38,590	17.1	PROV_L3_N
PROVIDER	PROVIDER_NPI_FLAG	6,618	38,590	17.1	PROV_L3_NPIFLAG
PROVIDER	PROVIDER_SEX	23,917	38,590	62.0	PROV_L3_SEX
PROVIDER	PROVIDER_SPECIALTY_PRIMARY	33,983	38,590	88.1	PROV_L3_SPECIALTY
PRO_CM	ENCOUNTERID	2,278,790	29,359,922	7.8	PROCM_L3_N
PRO_CM	PRO_CAT	29,359,922	29,359,922	100.0	PROCM_L3_CAT
PRO_CM	PRO_CODE	24,433,552	29,359,922	83.2	PROCM_L3_CODE_TYPE
PRO_CM	PRO_FULLNAME	0	29,359,922		PROCM_L3_FULLNAME
PRO_CM	PRO_METHOD	29,359,922	29,359,922	100.0	PROCM_L3_METHOD
PRO_CM	PRO_MODE	29,359,922	29,359,922	100.0	PROCM_L3_MODE
PRO_CM	PRO_NAME	0	29,359,922		PROCM_L3_NAME
PRO_CM	PRO_SOURCE	24,433,552	29,359,922	83.2	PROCM_L3_SOURCE
PRO_CM	PRO_TIME	0	29,359,922		XTBL_L3_TIMES
PRO_CM	PRO_TYPE	24,433,552	29,359,922	83.2	PROCM_L3_TYPE
VITAL	ENCOUNTERID	3,036,660	36,127,241	8.4	VIT_L3_N
VITAL	MEASURE_TIME	0	36,127,241		XTBL_L3_TIMES
VITAL	VITAL_SOURCE	0	36,127,241		VIT_L3_VITAL_SOURCE

Table excludes records with values outside of CDM specifications.

The four 'flavors of null' defined in the CDM are combined here but details are available in the source tables.

The denominator is derived from the applicable _N query (ALL_N + NULL_N for TAG=PATID).

Table IVE. Principal Diagnoses for Institutional Encounters

This table shows principal diagnosis code data availability for institutional encounters. Results support Data Check 3.06 (More than 10% of IP (inpatient) or ED to inpatient (EI) encounters with any diagnosis from a known DX_ORIGIN don't have a principal diagnosis from that source) and Data Check 2.07 (the average number of principal diagnoses per known DX_ORIGIN per encounter is above threshold [2.0 for inpatient (IP) and ED to inpatient (EI)]). For data check 3.06, exceptions are triggered when the percentage exceeds 10% or when 0 records have a principal diagnosis. Exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Encounter Type	DX_ORIGIN	Distinct encounter IDs with a principal diagnosis	Distinct encounter IDs without a principal diagnosis	% of encounters without a principal diagnosis	Principal diagnoses	Principal diagnoses per encounter with any principal diagnosis	Source table
EI (ED to IP Stay)	BI (billing)	176,107	0	0.0	177,956	1.0	DIA_L3_PDX_ENCTYPE
	CL (claim)	0	0		0		DIA_L3_PDXGRP_ENCTYPE
	DR (derived)	0	0		0		
	OD (Order/EHR)	99,484	0	0.0	169,002	1.7	
IP (Inpatient Hospital Stay)	BI (billing)	146,481	0	0.0	147,454	1.0	DIA_L3_PDX_ENCTYPE
	CL (claim)	0	0		0		DIA_L3_PDXGRP_ENCTYPE
	DR (derived)	0	0		0		
	OD (Order/EHR)	92,339	0	0.0	154,741	1.7	
IS (Non-acute Institutional Stay)	BI (billing)	0	0		0		DIA_L3_PDX_ENCTYPE
	CL (claim)	0	0		0		DIA_L3_PDXGRP_ENCTYPE
	DR (derived)	0	0		0		
	OD (Order/EHR)	0	0		0		
OS (Observation Stay)	BI (billing)	0	0		0		DIA_L3_PDX_ENCTYPE
	CL (claim)	0	0		0		DIA_L3_PDXGRP_ENCTYPE
	DR (derived)	0	0		0		
	OD (Order/EHR)	0	0		0		

Table excludes records with values outside of CDM specifications.

Percentage is calculated by dividing the number of encounters (i.e. distinct ENCOUNTERIDs) without a principal diagnosis by the total number of encounters in the DIAGNOSIS table.

The number of encounters without a principal diagnosis is obtained by identifying distinct encounterids which are not in a list of distinct encounterIDs where PDX=P.

Number of principal diagnoses per encounter with any principal diagnosis is calculated by dividing the number of principal diagnoses by the number of encounters with a principal diagnosis.

Table IVF. Data Latency and Completeness of Encounter, Diagnoses, and Procedures, Past 2 Years

This table includes ENCOUNTER, DIAGNOSIS, and PROCEDURES from the most recent 24 month period; month -0 is the month the data curation query was run. Data completeness is determined by comparing the actual volume to the expected volume in each month. Expected volume is determined by taking the average volume during the benchmark period of months -12 to month -23. Data completeness is reported as a percentage of the benchmark average. Temporal differences may be affected by data availability, ETL processes, date shifting, secular trends, and/or changes in data provenance.

These data support Data Check 3.07 (encounters, diagnoses, or procedures in an ambulatory (AV), telehealth (TH), emergency department (ED), ED to inpatient (EI), or inpatient (IP) setting are less than 75% complete two months prior to the current month). Data check exceptions occur if the month -2 result is <75% of the benchmark average or 0 records. Data check exceptions are highlighted in blue. Data check exceptions and unexpected results (e.g. significant discrepancies in data completeness between the tables) should be investigated and explained in the ETL ADD.

Month	Calendar Month	Ambulatory, Telehealth, ED, Inpatient or ED-to-Inpatient encounters		Ambulatory, Telehealth, ED, Inpatient or ED-to-Inpatient diagnoses		Ambulatory, Telehealth, ED, Inpatient or ED-to-Inpatient procedures	
		Records	Percent of benchmark average	Records	Percent of benchmark average	Records	Percent of benchmark average
Month -0	04/2026	37,249	20.5	30,056	4.7	44,396	6.6
Month -1	03/2026	186,025	102.3	731,546	115.2	737,494	109.6
Month -2	02/2026	176,616	97.2	721,949	113.7	721,626	107.3
Month -3	01/2026	170,009	93.5	710,711	111.9	709,681	105.5
Month -4	12/2025	176,389	97.0	715,846	112.7	726,690	108.0
Month -5	11/2025	157,187	86.5	635,073	100.0	646,320	96.1
Month -6	10/2025	194,785	107.2	767,132	120.8	793,202	117.9
Month -7	09/2025	179,037	98.5	718,520	113.1	734,669	109.2
Month -8	08/2025	168,613	92.8	678,174	106.8	690,701	102.7
Month -9	07/2025	168,676	92.8	672,183	105.8	684,850	101.8
Month -10	06/2025	153,714	84.6	609,990	96.0	627,454	93.3
Month -11	05/2025	161,233	88.7	628,821	99.0	650,875	96.8

Table IVF. Data Latency and Completeness of Encounter, Diagnoses, and Procedures, Past 2 Years (continued)

This table includes ENCOUNTER, DIAGNOSIS, and PROCEDURES from the most recent 24 month period; month -0 is the month the data curation query was run. Data completeness is determined by comparing the actual volume to the expected volume in each month. Expected volume is determined by taking the average volume during the benchmark period of months -12 to month -23. Data completeness is reported as a percentage of the benchmark average. Temporal differences may be affected by data availability, ETL processes, date shifting, secular trends, and/or changes in data provenance.

These data support Data Check 3.07 (encounters, diagnoses, or procedures in an ambulatory (AV), telehealth (TH), emergency department (ED), ED to inpatient (EI), or inpatient (IP) setting are less than 75% complete two months prior to the current month). Data check exceptions occur if the month -2 result is <75% of the benchmark average or 0 records. Data check exceptions are highlighted in blue. Data check exceptions and unexpected results (e.g. significant discrepancies in data completeness between the tables) should be investigated and explained in the ETL ADD.

		Ambulatory, ED, Inpatient or ED-to-Inpatient encounters		Ambulatory, ED, Inpatient or ED-to-Inpatient diagnoses		Ambulatory, ED, Inpatient or ED-to-Inpatient procedures	
		Records	Percent of benchmark average	Records	Percent of benchmark average	Records	Percent of benchmark average
Month	Calendar Month						
Benchmark Period							
Month -12	04/2025	171,704	94.5	660,917	104.1	674,653	100.3
Month -13	03/2025	181,015	99.6	685,941	108.0	706,756	105.1
Month -14	02/2025	167,925	92.4	617,322	97.2	636,987	94.7
Month -15	01/2025	178,144	98.0	658,692	103.7	676,851	100.6
Month -16	12/2024	179,702	98.9	638,212	100.5	667,722	99.3
Month -17	11/2024	179,046	98.5	618,752	97.4	651,632	96.9
Month -18	10/2024	208,582	114.8	702,195	110.5	759,528	112.9
Month -19	09/2024	182,391	100.3	622,288	98.0	674,906	100.3
Month -20	08/2024	188,778	103.9	617,835	97.3	674,129	100.2
Month -21	07/2024	185,962	102.3	621,921	97.9	671,283	99.8
Month -22	06/2024	168,222	92.6	562,290	88.5	608,433	90.5
Month -23	05/2024	189,626	104.3	615,866	97.0	668,791	99.4
Benchmark average		181,758		635,186		672,639	
Source table		ENC_L3_ENCTYPE_ADATE_YM		DIA_L3_ENCTYPE_ADATE_YM		PRO_L3_ENCTTPE_ADATE_YM	

Table IVG. Data Latency and Completeness of Vital, Prescribing, and Lab Data, Past 2 Years

This table includes VITAL, PRESCRIBING, and LAB_RESULT_CM data from the most recent 24 month period; month -0 is the month the data curation query was run. Data completeness is determined by comparing the actual volume to the expected volume in each month. Expected volume is determined by taking the average volume during the benchmark period of months -12 to month -23. Data completeness is reported as a percentage of the benchmark average. Temporal differences may be affected by data availability, ETL processes, date shifting, secular trends, and/or changes in data provenance.

These data support Data Check 3.11 (vital, prescribing, or laboratory records are less than 75% complete three months prior to the current month). Data check exceptions occur if the month -3 result is <75% of the benchmark average or 0 records. Data check exceptions are highlighted in blue. Data check exceptions and unexpected results should be investigated and explained in the ETL ADD.

Month	Calendar Month	Vitals		Prescriptions		Labs	
		Records	Percent of benchmark average	Records	Percent of benchmark average	Records	Percent of benchmark average
Month -0	04/2026	0		106,442	13.4	41	0.0
Month -1	03/2026	382,893	107.3	843,216	106.4	2,164,687	113.5
Month -2	02/2026	307,510	86.2	683,677	86.3	1,747,244	91.6
Month -3	01/2026	370,544	103.9	816,209	103.0	2,072,085	108.6
Month -4	12/2025	379,171	106.3	837,972	105.7	2,076,877	108.9
Month -5	11/2025	343,683	96.3	740,531	93.4	1,876,504	98.4
Month -6	10/2025	387,233	108.5	847,198	106.9	2,137,403	112.1
Month -7	09/2025	375,902	105.4	792,203	100.0	2,068,757	108.5
Month -8	08/2025	363,959	102.0	790,778	99.8	2,033,257	106.6
Month -9	07/2025	367,496	103.0	790,375	99.7	2,012,406	105.5
Month -10	06/2025	344,374	96.5	752,727	95.0	1,875,778	98.3
Month -11	05/2025	351,284	98.5	779,360	98.3	1,925,468	100.9

Table IVG. Data Latency and Completeness of Vital, Prescribing, and Lab Data, Past 2 Years (continued)

This table includes VITAL, PRESCRIBING, and LAB_RESULT_CM data from the most recent 24 month period; month -0 is the month the data curation query was run. Data completeness is determined by comparing the actual volume to the expected volume in each month. Expected volume is determined by taking the average volume during the benchmark period of months -12 to month -23. Data completeness is reported as a percentage of the benchmark average. Temporal differences may be affected by data availability, ETL processes, date shifting, secular trends, and/or changes in data provenance.

These data support Data Check 3.11 (vital, prescribing, or laboratory records are less than 75% complete three months prior to the current month). Data check exceptions occur if the month -3 result is <75% of the benchmark average or 0 records. Data check exceptions are highlighted in blue. Data check exceptions and unexpected results should be investigated and explained in the ETL ADD.

		Vitals		Prescriptions		Labs	
		Records	Percent of benchmark average	Records	Percent of benchmark average	Records	Percent of benchmark average
Month	Calendar Month						
Benchmark Period							
Month -12	04/2025	351,662	98.6	759,799	95.9	1,929,533	101.2
Month -13	03/2025	376,552	105.6	802,351	101.2	2,120,188	111.2
Month -14	02/2025	342,388	96.0	765,172	96.5	1,875,356	98.3
Month -15	01/2025	373,522	104.7	788,651	99.5	1,974,719	103.5
Month -16	12/2024	364,000	102.0	799,141	100.8	1,923,978	100.9
Month -17	11/2024	350,400	98.2	761,181	96.0	1,831,815	96.0
Month -18	10/2024	387,635	108.7	1,047,924	132.2	2,042,868	107.1
Month -19	09/2024	356,305	99.9	796,265	100.5	1,886,935	98.9
Month -20	08/2024	312,774	87.7	644,374	81.3	1,536,209	80.5
Month -21	07/2024	362,920	101.7	792,736	100.0	1,954,074	102.4
Month -22	06/2024	340,191	95.4	749,641	94.6	1,841,536	96.5
Month -23	05/2024	362,510	101.6	803,053	101.3	1,972,624	103.4
Benchmark average		356,738		792,524		1,907,486	
Source table		VIT_L3_MDATE_YM		PRES_L3_ODATE_YM		LAB_L3_RDATE_YM	

Table IVH. RXNORM Term Type Mapping

This table shows the number of records in the PRESCRIBING and MED_ADMIN tables by RXNORM Term Type tier. Guidance on mapping prescribing orders to RXNORM is provided in the CDM specifications. These data support Data Check 3.08 (less than 80% of prescribing orders are mapped to a RXCUI which fully specifies the ingredient, strength and dose form) and Data Check 3.15 (less than 80% of medication administrations mapped to RXNORM are mapped to a RXCUI that fully specifies the ingredient, strength and dose form). The numerator is the number of records per RXNORM tier; the denominator is the total number of records in the table for the specified time period. Data check exceptions occur if the Tier 1 percentage is <80% or the numerator is 0. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD. The percentage of prescribing orders mapped to Tier 1 brand drugs (a subset of the Tier 1 percentage) is also displayed.

Table	Term Type Tier	Term Type Tier Description	Term Types	Numerator	Percentage	Source table
PRESCRIBING	Tier 1	RXCUI encodes ingredient(s), strength and dose form	SCD, SBD, BPCCK, and GPCK	65,611,684	82.46	PRES_L3_RXCUI_TIER
	Tier 1 brand	RXCUI encodes ingredient(s), strength, dose form and brand	SBD, BPCCK	13,812,425	21.05	PRES_L3_RXCUI
	Tier 2	RXCUI encodes ingredient(s) and potentially strength or dose form. Can still represent medications with multiple ingredients with a single RXCUI.	SBDF, SBDFP, SCDF, SCDFP, SBDG, SCDG, SCDGP, SBDC, BN, and MIN	693,327	0.87	PRES_L3_RXCUI_TIER
	Tier 3	Requires more than one RXCUI to represent medications with multiple ingredients.	SCDC, PIN, and IN	891,541	1.12	PRES_L3_RXCUI_TIER
	Tier 4	RXCUI does not encode any ingredient information.	DF and DFG	143	0.00	PRES_L3_RXCUI_TIER
	Unknown	RXCUI was not populated or could not be matched to the reference table	n/a	12,374,507	15.55	PRES_L3_RXCUI_TIER

Term types were obtained from the rxnorm_cui_ref table. This table was created from the RxNorm 10/6/2025 Full Update Release. It includes all RXCUIs from the RXNORM terminology (i.e., SAB=RXNORM) with one of the following term types: BN (Brand Name), BPCCK (Brand Name Pack), DF (Dose Form), DFG (Dose Form Group), GPCK (Generic Pack), IN (Ingredient), MIN (Multiple Ingredients), SBD (Semantic Branded Drug), SBDC (Semantic Branded Drug Component), SBDF (Semantic Branded Drug Form), SBDFP (Semantic Branded Drug Form Precise), SBDG (Semantic Branded Dose Form Group), SCD (Semantic Clinical Drug), SCDC (Semantic Clinical Drug Component), SCDF (Semantic Clinical Drug Form), SCDFP (Semantic Clinical Drug Form Precise), SCDG (Semantic Clinical Dose Form Group), SCDGP (Semantic Clinical Dose Form Group Precise), or PIN (Precise Ingredient).

Table IVI. Laboratory and Clinical Observation Result Data Completeness

This table shows the level of data completeness for LAB_RESULT_CM and OBS_CLIN records and supports Data Check 3.09 (less than 80% of laboratory results are mapped to LAB_LOINC), Data Check 3.10 (less than 80% of quantitative results for tests mapped to LAB_LOINC fully specify the normal range), Data Check 3.12 (less than 80% of quantitative results for tests mapped to LAB_LOINC fully specify the result unit), Data Check 3.16 (less than 80% of clinical observations are mapped to an OBSCLIN_CODE and have a quantitative, qualitative, or narrative result), and Data Check 3.17 (less than 80% of quantitative results for tests mapped to OBSCLIN_CODE fully specify the RESULT_UNIT). Data check exceptions occur if the percentages are <80% or the numerator is 0. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Data Check	Description	Numerator	Denominator	Percentage	Source table
LAB_RESULT_CM	n/a	Number of distinct LAB_LOINCs	1,725			LAB_L3_LOINC
	n/a	Results mapped to a known LAB_LOINC	150,301,142	167,597,234	89.68	LAB_L3_RECORDC; LAB_L3_N
	3.09	Results mapped to a known LAB_LOINC with a known result	142,181,092	167,597,234	84.83	LAB_L3_RECORDC
	n/a	Quantitative results	129,673,774			LAB_L3_RECORDC
	3.10	Quantitative results which fully specify the normal range	111,332,945	129,673,774	85.86	
	3.12	Quantitative results which specify the result unit	114,064,495	129,673,774	87.96	
OBS_CLIN	n/a	Number of distinct OBSCLIN_CODEs	232			OBSCLIN_L3_CODE_ TYPE
	n/a	Results mapped to a known OBSCLIN_CODE	486,423,175	486,423,175	100.00	OBSCLIN_L3_ RECORDC; OBSCLIN_L3_N
	3.16	Results mapped to a known OBSCLIN_CODE with a known result	486,423,175	486,423,175	100.00	OBSCLIN_L3_ RECORDC; OBSCLIN_L3_N
	n/a	Quantitative results	277,492,492			OBSCLIN_L3_ RECORDC
	3.17	Quantitative results which specify the result unit	240,416,171	277,492,492	86.64	OBSCLIN_L3_ RECORDC

The denominator for the percentage of results mapped to a known LAB_LOINC or OBSCLIN_CODE is the total number of records in the table (ALL_N + NULL_N for the LAB_RESULT_CM_ID or OBSCLIN_ID field). The criteria for each measurement are described in table IVI_REF.

Table IVI_Ref. Laboratory and Clinical Observation Result Data Completeness Definitions

This table shows the definitions for each of the numerator values in Table IVI.

Description	Source Table Row	CODE criteria	RESULT_NUM criteria	RESULT_MODIFIER criteria	Other criteria
Number of distinct codes		Not null			
Results mapped to a known code	KNOWN_TEST	Not null			
Results mapped to a known code with a known result	KNOWN_TEST_RESULT	Not null			[RESULT_NUM is not null and RESULT_MODIFIER is not in (NI, UN, OT, null)] or RESULT_QUAL is not in (NI, UN, OT, null) or RESULT_TEXT is not null
Quantitative results	KNOWN_TEST_RESULT_NUM	Not null	Not null	not in (NI, UN, OT, null)	
Quantitative results which specify the result unit	KNOWN_TEST_RESULT_NUM_UNIT	Not null	Not null	not in (NI, UN, OT, null)	RESULT_UNIT is not in ('NI','UN','OT', null)
Quantitative results which fully specify the normal range.	KNOWN_TEST_NUM_RESULT_RANGE	Not null	Not null	not in (NI, UN, OT, null)	[NORM_MODIFIER_LOW='EQ' and NORM_MODIFIER_HIGH='EQ' and NORM_RANGE_LOW is not null and NORM_RANGE_HIGH is not null] or [NORM_MODIFIER_LOW in ('GT','GE') and NORM_MODIFIER_HIGH='NO' and NORM_RANGE_LOW is not null and NORM_RANGE_HIGH is null] or [NORM_MODIFIER_HIGH in ('LE','LT') and NORM_MODIFIER_LOW='NO' and NORM_RANGE_HIGH is not null and NORM_RANGE_LOW is null].

Table IVJ. Data Latency and Completeness of Medication Administration, Dispensing and Clinical Observation Data, Past 2 Years

This table includes MED_ADMIN, DISPENSING, and OBS_CLIN data from the most recent 24 month period; month -0 is the month the data curation query was run. Data completeness is determined by comparing the actual volume to the expected volume in each month. Expected volume is determined by taking the average volume during the benchmark period of months -12 to month -23. Data completeness is reported as a percentage of the benchmark average. Temporal differences may be affected by data availability, ETL processes, date shifting, secular trends, and/or changes in data provenance.

These data support Data Check 3.14 (medication administration, dispensing, or clinical observation records are less than 75% complete three months prior to the current month). Data check exceptions occur if the month -3 result is <75% of the benchmark average or 0 records. Data check exceptions are highlighted in blue. Data check exceptions and unexpected results should be investigated and explained in the ETL ADD.

Month	Calendar Month	MED_ADMIN		DISPENSING		OBS_CLIN	
		Records	Percent of benchmark average	Records	Percent of benchmark average	Records	Percent of benchmark average
Month -0	04/2026	0		4,552	10.1	39	0.0
Month -1	03/2026	438,257	107.3	54,007	120.1	5,224,334	109.4
Month -2	02/2026	323,239	79.1	49,914	111.0	4,239,629	88.8
Month -3	01/2026	443,005	108.4	50,317	111.9	5,039,902	105.5
Month -4	12/2025	434,538	106.4	57,420	127.7	5,204,831	109.0
Month -5	11/2025	410,660	100.5	46,964	104.4	4,623,848	96.8
Month -6	10/2025	446,100	109.2	56,357	125.3	5,258,953	110.1
Month -7	09/2025	429,400	105.1	57,766	128.4	5,028,446	105.3
Month -8	08/2025	435,475	106.6	46,576	103.6	4,929,215	103.2
Month -9	07/2025	428,769	105.0	46,749	103.9	4,988,590	104.4
Month -10	06/2025	407,941	99.9	44,536	99.0	4,649,978	97.4
Month -11	05/2025	421,342	103.1	45,227	100.6	4,739,996	99.2

Table IVJ. Data Latency and Completeness of Medication Administration, Dispensing and Clinical Observation Data, Past 2 Years
(continued)

This table includes MED_ADMIN, DISPENSING, and OBS_CLIN data from the most recent 24 month period; month -0 is the month the data curation query was run. Data completeness is determined by comparing the actual volume to the expected volume in each month. Expected volume is determined by taking the average volume during the benchmark period of months -12 to month -23. Data completeness is reported as a percentage of the benchmark average. Temporal differences may be affected by data availability, ETL processes, date shifting, secular trends, and/or changes in data provenance.

These data support Data Check 3.14 (medication administration, dispensing, or clinical observation records are less than 75% complete three months prior to the current month). Data check exceptions occur if the month -3 result is <75% of the benchmark average or 0 records. Data check exceptions are highlighted in blue. Data check exceptions and unexpected results should be investigated and explained in the ETL ADD.

		MED_ADMIN		DISPENSING		OBS_CLIN	
		Records	Percent of benchmark average	Records	Percent of benchmark average	Records	Percent of benchmark average
Month	Calendar Month						
Benchmark Period							
Month -12	04/2025	396,390	97.0	46,087	102.5	4,737,832	99.2
Month -13	03/2025	436,017	106.7	45,681	101.6	5,056,788	105.9
Month -14	02/2025	409,733	100.3	42,876	95.3	4,486,543	93.9
Month -15	01/2025	428,577	104.9	46,251	102.8	4,879,102	102.2
Month -16	12/2024	419,662	102.7	46,952	104.4	4,820,683	100.9
Month -17	11/2024	412,058	100.9	42,398	94.3	4,716,026	98.7
Month -18	10/2024	433,055	106.0	49,126	109.2	5,182,465	108.5
Month -19	09/2024	411,350	100.7	44,228	98.3	4,768,024	99.8
Month -20	08/2024	330,020	80.8	44,516	99.0	4,196,453	87.9
Month -21	07/2024	410,856	100.6	45,658	101.5	4,928,116	103.2
Month -22	06/2024	405,310	99.2	40,892	90.9	4,606,391	96.4
Month -23	05/2024	409,020	100.1	45,046	100.2	4,935,401	103.3
Benchmark average		408,504		44,976		4,776,152	
Source table		MEDADM_L3_ SDATE_YM		DISP_L3_ DDATE_YM		OBSCLIN_L3_ SDATE_YM	

Table VA. Changes in Tables

This table shows changes in key DataMart attributes between the most recent approved DataMart refresh and the current DataMart refresh and supports Data Check 4.01 (more than a 5% decrease in the number of patients or records in a CDM table). Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Table	Records			Patients			Source table
	Previous Refresh	Current Refresh	Percent Change	Previous Refresh	Current Refresh	Percent Change	
DEMOGRAPHIC	2,275,381	2,284,557	0.4	2,275,381	2,284,557	0.4	DEM_L3_N
ENROLLMENT	817,717	830,128	1.5	817,717	830,128	1.5	ENR_L3_N
ENCOUNTER	15,107,679	15,658,577	3.6	817,204	829,623	1.5	ENC_L3_N
DIAGNOSIS	51,416,445	53,705,073	4.5	800,079	812,736	1.6	DIA_L3_N
PROCEDURES	56,488,685	58,774,019	4.0	796,051	808,780	1.6	PRO_L3_N
VITAL	35,056,546	36,127,241	3.1	850,331	861,005	1.3	VIT_L3_N
DEATH	452,816	454,934	0.5	371,874	373,556	0.5	DEATH_L3_N
PRESCRIBING	76,986,677	79,571,202	3.4	750,147	760,185	1.3	PRES_L3_N
DISPENSING	4,472,370	4,824,060	7.9	236,885	252,357	6.5	DISP_L3_N
LAB_RESULT_CM	161,551,989	167,597,234	3.7	530,169	541,068	2.1	LAB_L3_N
CONDITION	2,738,413	2,791,908	2.0	471,408	477,354	1.3	COND_L3_N
DEATH_CAUSE	18,654	19,017	1.9	18,243	18,601	2.0	DEATHC_L3_N
PRO_CM	28,464,236	29,359,922	3.1	746,406	756,435	1.3	PROCM_L3_N
PROVIDER	38,448	38,590	0.4	---	0		PROV_L3_N
MED_ADMIN	37,267,189	38,479,890	3.3	503,391	513,380	2.0	MEDADM_L3_N
OBS_CLIN	465,320,070	486,423,175	4.5	849,418	862,879	1.6	OBSCLIN_L3_N
OBS_GEN	83,900,752	87,218,390	4.0	826,581	837,440	1.3	OBSGEN_L3_N
HASH_TOKEN	2,275,381	2,284,557	0.4	2,275,381	2,284,557	0.4	HASH_L3_N
IMMUNIZATION	2,379,275	2,436,968	2.4	370,646	375,540	1.3	IMMUNE_L3_N
LDS_ADDRESS_HISTORY	2,842,736	2,852,571	0.3	2,198,272	2,207,036	0.4	LDSADRS_L3_N
LAB_HISTORY	---	0		---	0		LABHIST_L3_N
EXTERNAL_MEDS	4,497,906	4,599,717	2.3	543,776	551,286	1.4	EXTMED_L3_N
PAT_RELATIONSHIP	80,853	82,582	2.1	80,853	82,582	2.1	PATREL_L3_N

Percent change is calculated as current minus previous divided by previous multiplied by 100.

Table VB. Changes in Selected Encounter Types and Domains

This table shows changes in key DataMart attributes between the most recent approved DataMart refresh and the current DataMart refresh and supports Data Check 4.02 [more than a 5% decrease in the number of patients or records for diagnosis, procedures, labs or prescriptions during an ambulatory (AV), telehealth (TH), other ambulatory (OA), emergency department (ED), or inpatient (IP) encounter]. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

Encounter Type	Records			Patients			Source table
	Previous Refresh	Current Refresh	Percent Change	Previous Refresh	Current Refresh	Percent Change	
Ambulatory Visit (AV)							
Diagnoses	37,599,472	39,314,798	4.6	734,084	745,742	1.6	DIA_L3_ENCTYPE
Procedures	38,304,672	39,945,453	4.3	731,110	742,885	1.6	PRO_L3_ENCTYPE
Labs	67,246,683	69,989,632	4.1	435,166	444,606	2.2	XTBL_L3_LAB_ENCTYPE
Prescriptions	23,735,961	24,617,319	3.7	549,929	560,172	1.9	XTBL_L3_PRES_ENCTYPE
Emergency Department (ED)							
Diagnoses	4,164,599	4,348,678	4.4	270,260	277,257	2.6	DIA_L3_ENCTYPE
Procedures	4,889,247	5,076,370	3.8	259,888	266,713	2.6	PRO_L3_ENCTYPE
Labs	16,260,628	16,876,130	3.8	161,423	166,102	2.9	XTBL_L3_LAB_ENCTYPE
Prescriptions	2,494,887	2,580,441	3.4	208,668	214,211	2.7	XTBL_L3_PRES_ENCTYPE
Inpatient (IP)							
Diagnoses	3,099,181	3,209,018	3.5	108,464	111,449	2.8	DIA_L3_ENCTYPE
Procedures	5,281,112	5,436,994	3.0	108,353	111,335	2.8	PRO_L3_ENCTYPE
Labs	29,150,086	29,948,010	2.7	105,199	107,872	2.5	XTBL_L3_LAB_ENCTYPE
Prescriptions	17,813,104	18,315,348	2.8	108,165	110,943	2.6	XTBL_L3_PRES_ENCTYPE
Other Ambulatory (OA)							
Diagnoses	0	0		0	0		DIA_L3_ENCTYPE
Procedures	0	0		0	0		PRO_L3_ENCTYPE
Labs	0	0		0	0		XTBL_L3_LAB_ENCTYPE
Prescriptions	0	0		0	0		XTBL_L3_PRES_ENCTYPE
Telehealth (TH) encounters							
Diagnoses	1,229,150	1,286,087	4.6	110,741	113,896	2.8	DIA_L3_ENCTYPE
Procedures	572,066	595,607	4.1	106,997	110,262	3.1	PRO_L3_ENCTYPE
Labs	64,077	64,827	1.2	2,970	3,016	1.5	XTBL_L3_LAB_ENCTYPE
Prescriptions	449,019	468,938	4.4	62,592	64,506	3.1	XTBL_L3_PRES_ENCTYPE

Percent change is calculated as current minus previous divided by previous multiplied by 100.

Table VC. Changes in Selected Code Types

This table shows changes in key DataMart attributes between the most recent approved DataMart refresh and the current DataMart refresh and supports Data Check 4.03 (more than a 5% decrease in the number of records or distinct codes for CPT/HCPCS, CVX, ICD10, NDC, LOINC or RXCUI codes). The data check is not applied to ICD9 codes because these codes will decrease between refreshes because of the 10 year lookback. Data check exceptions are highlighted in blue and should be investigated and explained in the ETL ADD.

	Records			Distinct Codes			Source table
	Previous Refresh	Current Refresh	Percent Change	Previous Refresh	Current Refresh	Percent Change	
DIAGNOSIS							
09	62	63	1.6	43	44	2.3	DIA_L3_DX_DXTYPE
10	51,390,010	53,672,864	4.4	44,716	45,061	0.8	DIA_L3_DX_DXTYPE
PROCEDURES							
09	---	0		---	0		PRO_L3_PX_PXTYPE
10	520,217	536,961	3.2	14,310	14,443	0.9	PRO_L3_PX_PXTYPE
CH	55,937,134	58,204,604	4.1	11,412	11,569	1.4	PRO_L3_PX_PXTYPE
ND	---	0		---	0		PRO_L3_PX_PXTYPE
DISPENSING							
ND	4,472,370	4,824,060	7.9	22,220	24,107	8.5	DISP_L3_NDC
IMMUNIZATION							
CH	---	0		---	0		IMMUNE_L3_CODE_CODETYPE
CX	2,251,367	2,304,384	2.4	103	103	0.0	IMMUNE_L3_CODE_CODETYPE
ND	---	0		---	0		IMMUNE_L3_CODE_CODETYPE
RX	---	0		---	0		IMMUNE_L3_CODE_CODETYPE
MED_ADMIN							
ND	35,091,674	36,236,300	3.3	1,354	1,356	0.1	MEDADM_L3_CODE_TYPE
RX	---	0		---	0		MEDADM_L3_CODE_TYPE
PRESCRIBING							
RX	76,228,890	78,601,310	3.1	42,907	43,046	0.3	PRES_L3_RXCUI
OBS_CLIN							
LC	465,320,070	486,423,175	4.5	228	232	1.8	OBSCLIN_L3_CODE_TYPE
PRO_CM							
LC	4,747,369	4,926,370	3.8	58	58	0.0	PROCM_L3_CODE_TYPE
EXTERNAL_MEDS							
RX	4,102,235	4,186,887	2.1	25,403	25,548	0.6	EXTMED_L3_RXCUI

Percent change is calculated as current minus previous divided by previous multiplied by 100.